

SOC HOME LAB

INCIDENT INVESTIGATION REPORT

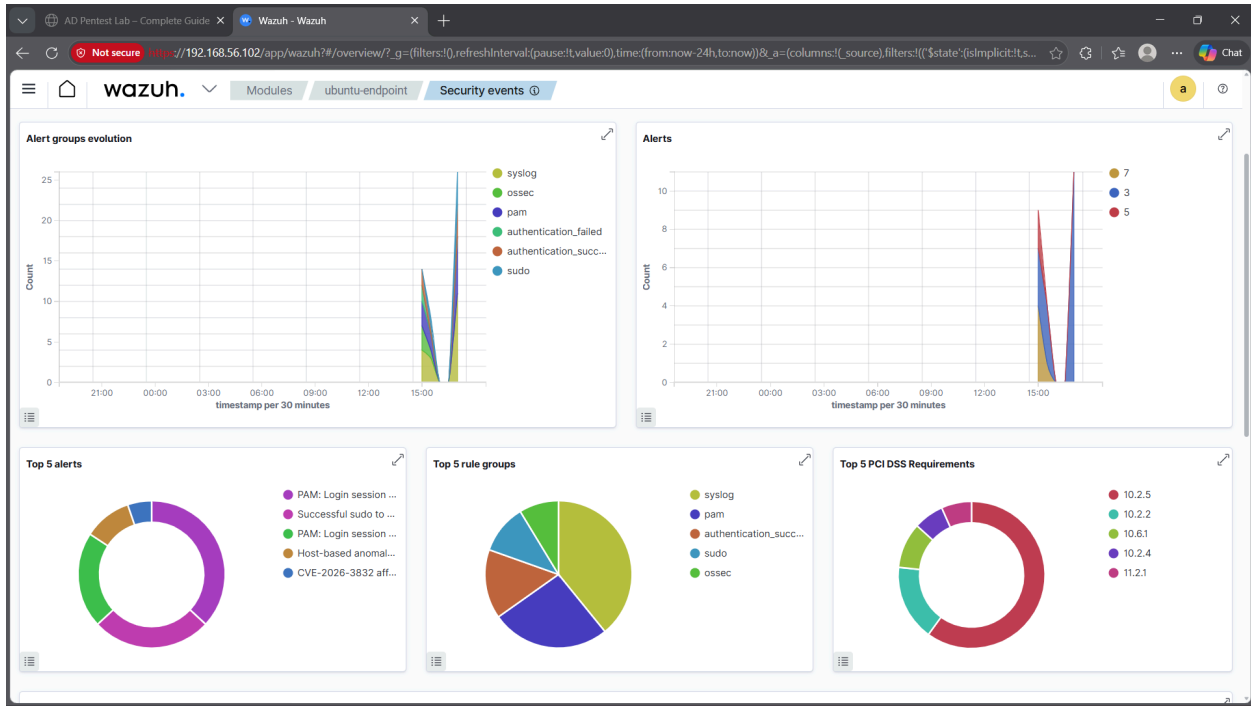


Figure: Wazuh Security Events overview showing alert activity from the lab environment

Field	Value
Analyst Name	Jazz
Report Date	05/05/2026
Lab Environment	Wazuh 4.7.5 all-in-one server, Windows 10 agent, Ubuntu 22.04.5 agent
Incident ID	INC-WAZUH-HOMELAB-001
Severity	High — multiple Level 10+ alerts observed
Status	Investigated — True positives confirmed in controlled lab

Executive Summary

This report documents a Wazuh SIEM/XDR home SOC lab built with one Wazuh server, one Windows endpoint, and one Ubuntu endpoint. The lab was used to simulate realistic security events including suspicious PowerShell execution, SSH login attempts, Linux sudo privilege escalation, file integrity changes, and vulnerability scanning. Wazuh successfully collected logs from both endpoints, generated alerts, mapped activity to MITRE ATT&CK, and supported analyst triage through event details, raw logs, severity levels, and detection rules.

Overall result: the lab is operational. Windows and Ubuntu agents are active, alerts are appearing in Security Events, MITRE ATT&CK mappings are visible, and Vulnerability Detection found CVEs on both Ubuntu and Windows endpoints.

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SECTION 1 — LAB SETUP & ARCHITECTURE

1.1 What is Wazuh?

Wazuh is a free, open-source **SIEM (Security Information and Event Management)** and **XDR (Extended Detection and Response)** platform. It collects, aggregates, and analyses security logs from endpoints in real time. It supports Windows and Linux agents, provides built-in detection rules, maps alerts to the MITRE ATT&CK framework, and exposes a REST API — making it ideal for a home SOC lab that mirrors enterprise tooling.

1.2 Lab Architecture

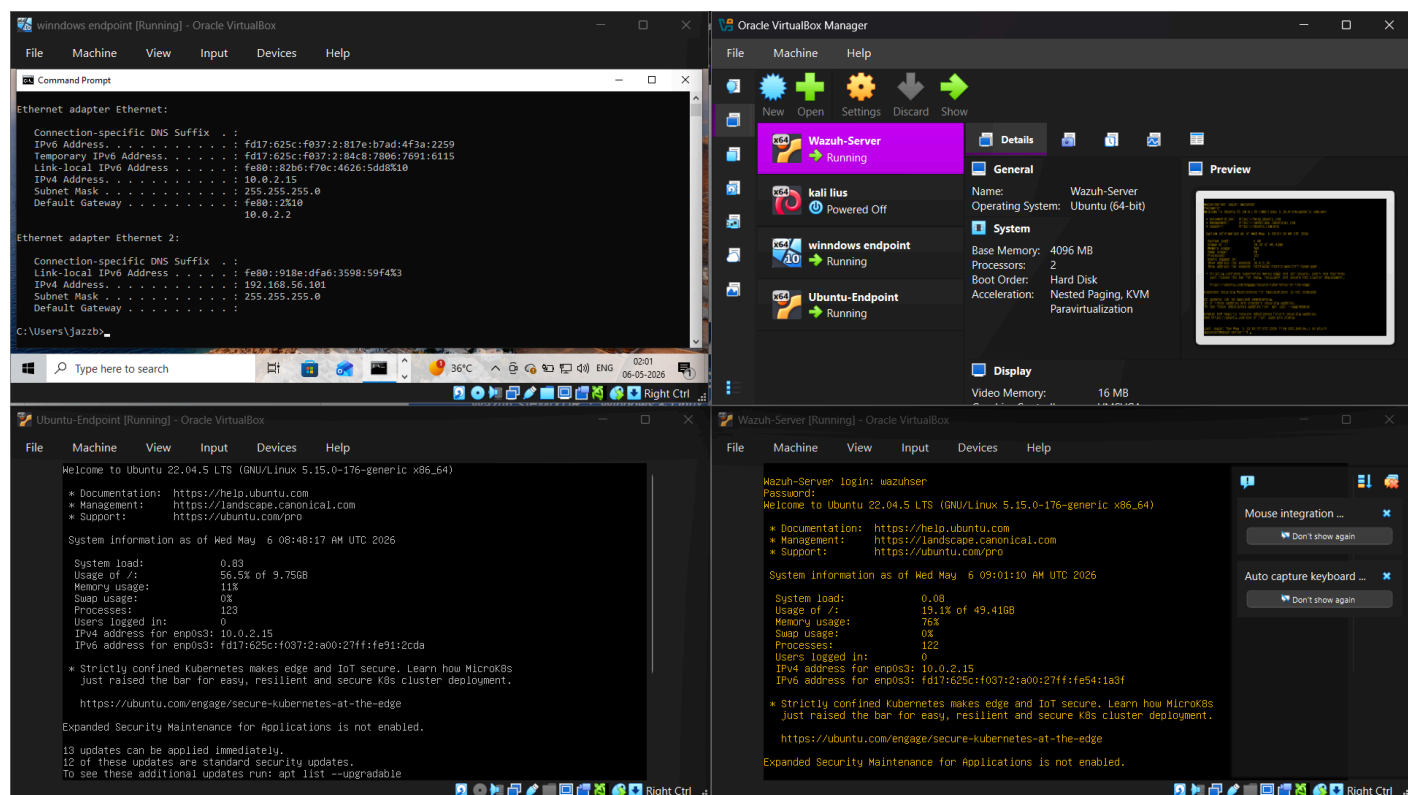


Figure: Lab Architecture

Component	OS / Version	Role	IP Address	Notes
Wazuh Server	Ubuntu Server 22.04 LTS	SIEM / Manager / Dashboard / Indexer	192.168.56.102	Wazuh 4.7.5
Windows Agent	Windows 10 Home 10.0.19045.6466	Endpoint / Windows logs / Sysmon source	192.168.56.101	Agent v4.7.0
Linux Agent	Ubuntu 22.04.5 LTS	Endpoint / Linux auth logs / FIM source	192.168.56.103	Agent v4.7.5
Attacker/Test Source	Ubuntu Endpoint and Windows VM; no dedicated Kali VM was used in final testing.	Controlled attack simulation	Host-only network	Hydra, SSH, PowerShell, sudo tests

The environment used a VirtualBox host-only network in the 192.168.56.0/24 range for VM-to-VM and host-to-VM communication. NAT was also used for internet access during package installation and vulnerability feed updates.

```
Wazuh-Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
[12420.907373] watchdog: BUG: soft lockup - CPU#0 stuck for 1751s! [swapper/0:0]
[20914.398672] watchdog: BUG: soft lockup - CPU#0 stuck for 691s! [swapper/0:0]
[20914.488826] systemd[1]: systemd-timesyncd.service: Watchdog timeout (limit 3min)!
[20914.489715] systemd[1]: systemd-logind.service: Watchdog timeout (limit 3min)!
[20914.542649] systemd[1]: snapd.service: Watchdog timeout (limit 5min)!
[20914.628423] systemd[1]: systemd-networkd.service: Watchdog timeout (limit 3min)!
[33151.894011] watchdog: BUG: soft lockup - CPU#0 stuck for 9712s! [swapper/0:0]
[33151.893732] rcu: INFO: rcu_sched self-detected stall on CPU
[33151.925475] rcu: #1-...!: (1 GPs behind) idle=d2f/0/0x1 softirq=207563/207563 fqs=0
[33151.926214] rcu: rcu_sched kthread timer wakeup didn't happen for 2606380 jiffies! g506221 f0x0 R
CU_GP_WAIT_FQS(5) ->state=0x402
[33151.926217] rcu: #Possible timer handling issue on cpu=0 timer-softirq=2433186
[33151.926218] rcu: rcu_sched kthread starved for 2606381 jiffies! g506221 f0x0 RCU_GP_WAIT_FQS(5) -
>state=0x402 ->cpu=0
[33151.926220] rcu: #Unless rcu_sched kthread gets sufficient CPU time, OOM is now expected behavior
.
[33151.926220] rcu: RCU grace-period kthread stack dump:
[33151.944362] rcu: Stack dump where RCU GP kthread last ran:

wazuhser@Wazuh-Server:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=128 time=2.17 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=128 time=0.933 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=128 time=0.824 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=128 time=2.26 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=128 time=0.954 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=128 time=0.891 ms
^C
--- 192.168.56.101 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5500ms
rtt min/avg/max/mdev = 0.824/1.338/2.255/0.620 ms
wazuhser@Wazuh-Server:~$ ip -br a
lo                UNKNOWN        127.0.0.1/8 ::1/128
enp0s3            UP                10.0.2.15/24 metric 100 fd17:625c:f037:2:a00:27ff:fe54:1a3f/64 fe80:
:a00:27ff:fe54:1a3f/64
enp0s8            UP                192.168.56.102/24 fe80::a00:27ff:fed8:e0fd/64
wazuhser@Wazuh-Server:~$ _
```

Figure: Wazuh Server's IP Address

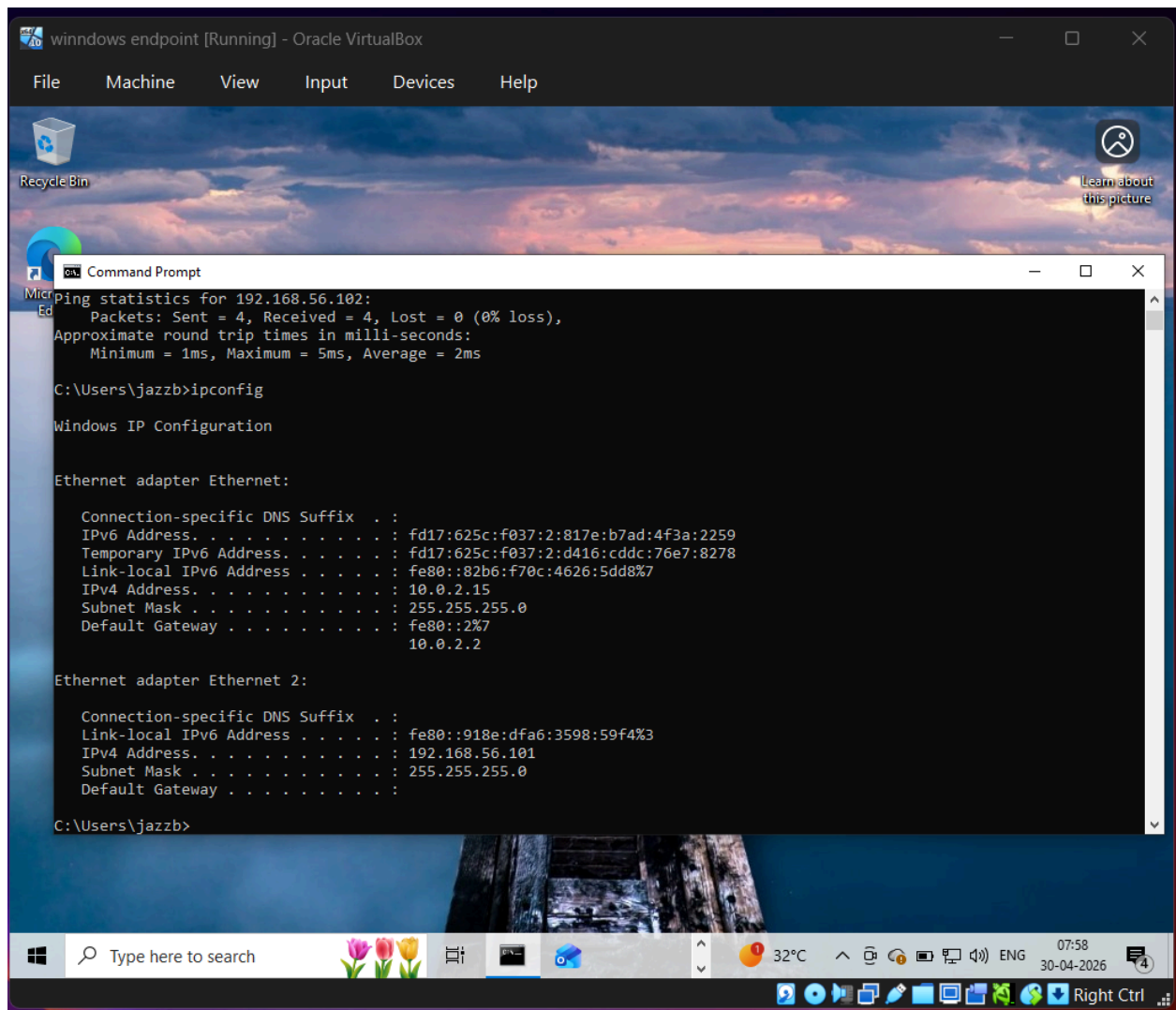


Figure: Windows' IP Address

```
Ubuntu-Endpoint [Running] - Oracle VirtualBox
File Machine View Input Devices Help
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=2.72 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.29 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.853 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=1.26 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.958 ms
^C
--- 192.168.56.102 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4632ms
rtt min/avg/max/mdev = 0.853/1.416/2.720/0.673 ms
jazz@ubuntu-endpoint:~$ ping 192.162.56.101
PING 192.162.56.101 (192.162.56.101) 56(84) bytes of data.
^C
--- 192.162.56.101 ping statistics ---
35 packets transmitted, 0 received, 100% packet loss, time 34942ms

jazz@ubuntu-endpoint:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=128 time=2.90 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=128 time=1.55 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=128 time=1.93 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=128 time=1.96 ms
^C
--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3286ms
rtt min/avg/max/mdev = 1.548/2.082/2.895/0.496 ms
jazz@ubuntu-endpoint:~$ ip -ba a
Cannot open file "a" for reading: No such file or directory
jazz@ubuntu-endpoint:~$ ls
jazz@ubuntu-endpoint:~$ ip -br a
lo                UNKNOWN          127.0.0.1/8 ::1/128
enp0s3            UP               10.0.2.15/24 metric 100 fd17:625c:f037:2:a00:27ff:fe91:2cda/64 fe80:
:a00:27ff:fe91:2cda/64
enp0s8            UP               192.168.56.103/24 fe80::a00:27ff:fe39:550f/64
jazz@ubuntu-endpoint:~$ _
```

Figure: Ubuntu's IP Address

1.3 Wazuh Dashboard — First Look

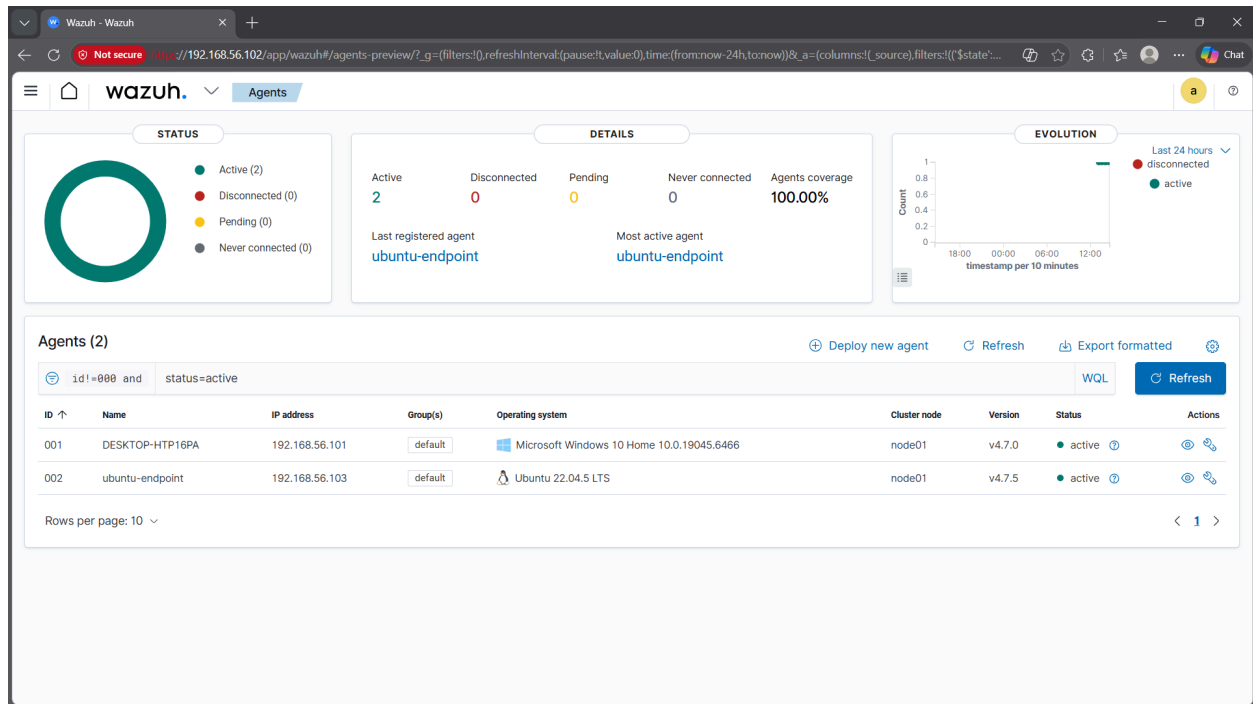


Figure: Two Agents on Wazuh Dashboard

1.4 Agent Installation Proof

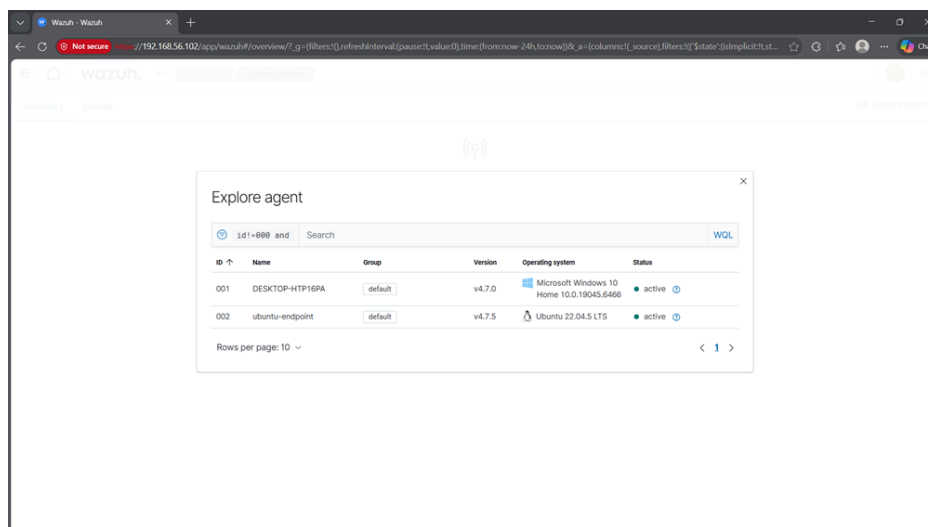


Figure: Wazuh Explore Agent view showing Windows and Ubuntu agents active

The Wazuh dashboard confirms two active endpoint agents: DESKTOP-HTP16PA for Windows and ubuntu-endpoint for Ubuntu. This proves that both endpoints are registered with the Wazuh Manager and are sending telemetry.

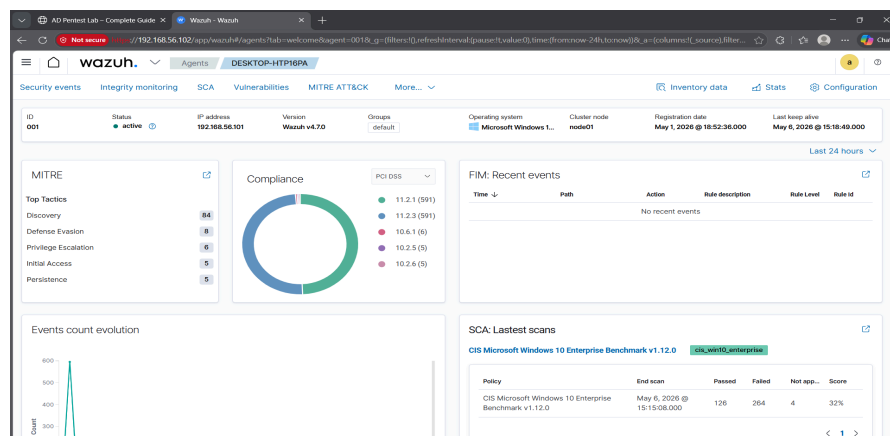


Figure: Windows Agent on Wazuh Dashboard

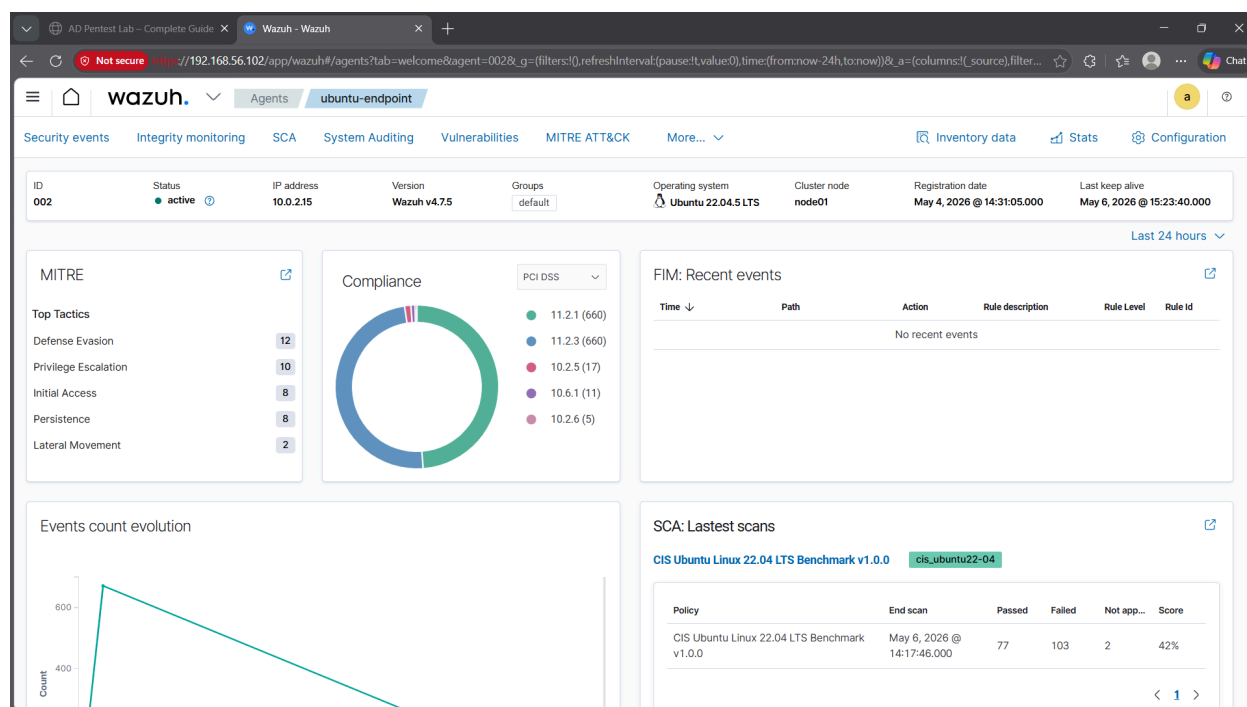


Figure: Ubuntu Agent on Wazuh Dashboard

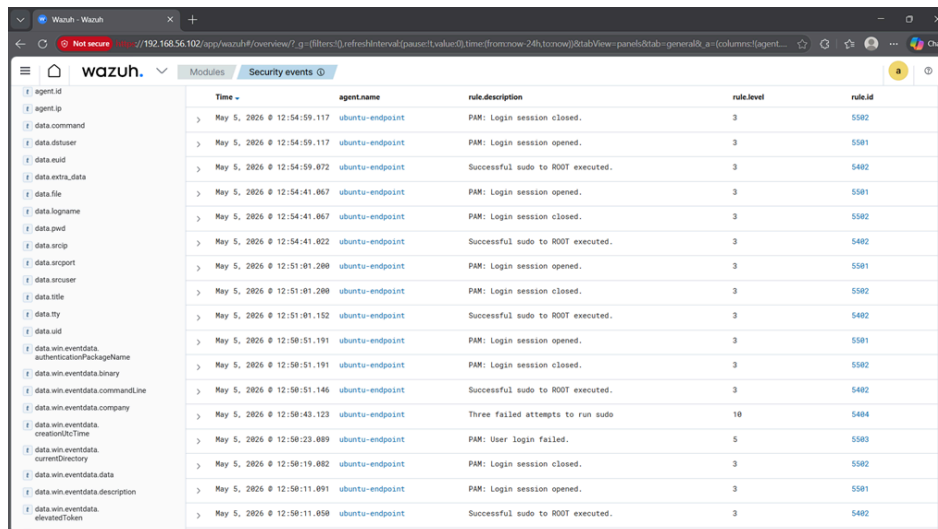
SECTION 2 — LOG COLLECTION & EVENT MONITORING

2.1 What Logs Wazuh Collects

Wazuh agents collect and forward the following log sources from each endpoint to the Wazuh Manager for correlation and alerting:

Log Source	Platform	What It Captures
Windows Security Event Log	Windows	Logon success/failure, user activity, account events
Windows Sysmon Operational	Windows	Process creation, suspicious processes, file drops, PowerShell behavior
Linux /var/log/auth.log	Ubuntu	SSH logins, sudo activity, PAM events, privilege escalation attempts
Linux Syslog	Ubuntu	Service messages, daemon activity, system events
File Integrity Monitoring / Syscheck	Both	File creation, deletion, modification, hash changes
Syscollector Inventory	Both	OS, packages/software, processes, ports, hardware data

2.2 Live Events Stream



Time	agent.name	rule.description	rule.level	rule.id
May 5, 2026 @ 12:54:59.117	ubuntu-endpoint	PAM: Login session closed.	3	5592
May 5, 2026 @ 12:54:59.117	ubuntu-endpoint	PAM: Login session opened.	3	5591
May 5, 2026 @ 12:54:59.872	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5492
May 5, 2026 @ 12:54:41.867	ubuntu-endpoint	PAM: Login session opened.	3	5591
May 5, 2026 @ 12:54:41.867	ubuntu-endpoint	PAM: Login session closed.	3	5592
May 5, 2026 @ 12:54:41.822	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5492
May 5, 2026 @ 12:51:01.289	ubuntu-endpoint	PAM: Login session opened.	3	5591
May 5, 2026 @ 12:51:01.289	ubuntu-endpoint	PAM: Login session closed.	3	5592
May 5, 2026 @ 12:51:01.152	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5492
May 5, 2026 @ 12:58:51.191	ubuntu-endpoint	PAM: Login session opened.	3	5591
May 5, 2026 @ 12:58:51.191	ubuntu-endpoint	PAM: Login session closed.	3	5592
May 5, 2026 @ 12:58:51.146	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5492
May 5, 2026 @ 12:58:43.123	ubuntu-endpoint	Three failed attempts to run sudo	10	5494
May 5, 2026 @ 12:58:23.899	ubuntu-endpoint	PAM: User login failed.	5	5593
May 5, 2026 @ 12:58:19.882	ubuntu-endpoint	PAM: Login session closed.	3	5592
May 5, 2026 @ 12:58:11.891	ubuntu-endpoint	PAM: Login session opened.	3	5591
May 5, 2026 @ 12:58:11.858	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5492

Figure: Security Events table showing events from Windows and Ubuntu endpoints

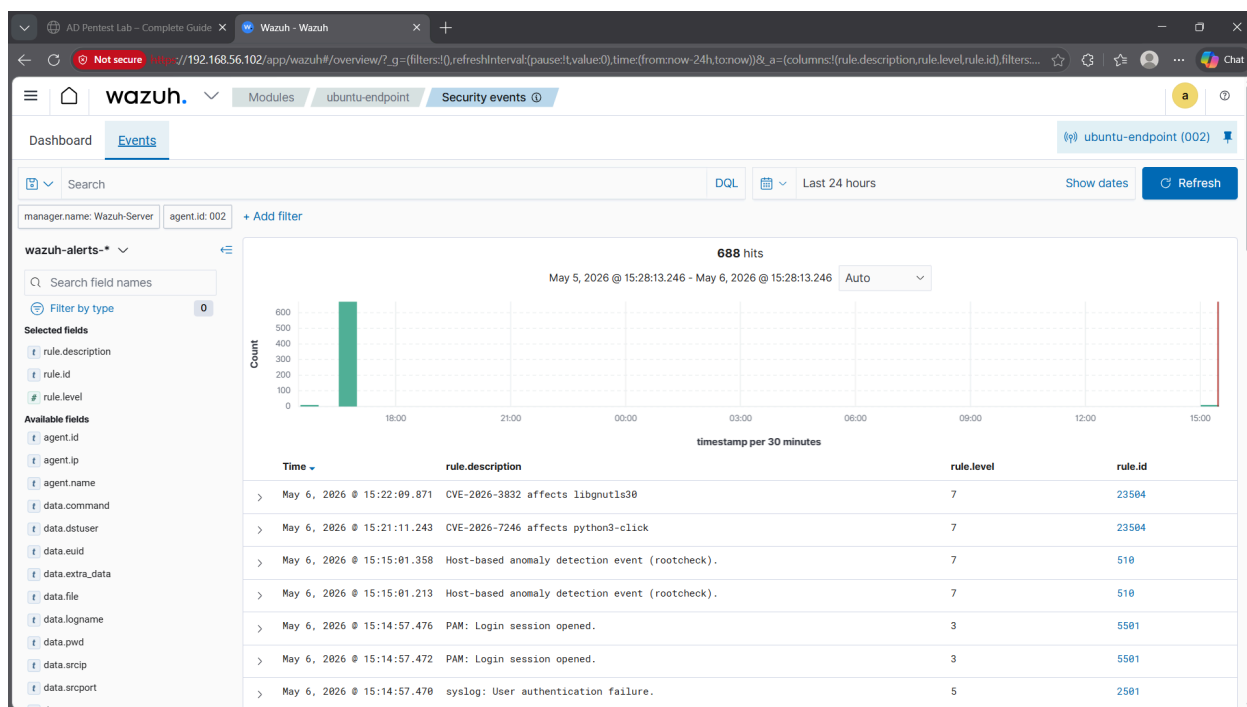


Figure: Wazuh Event Stream of Ubuntu

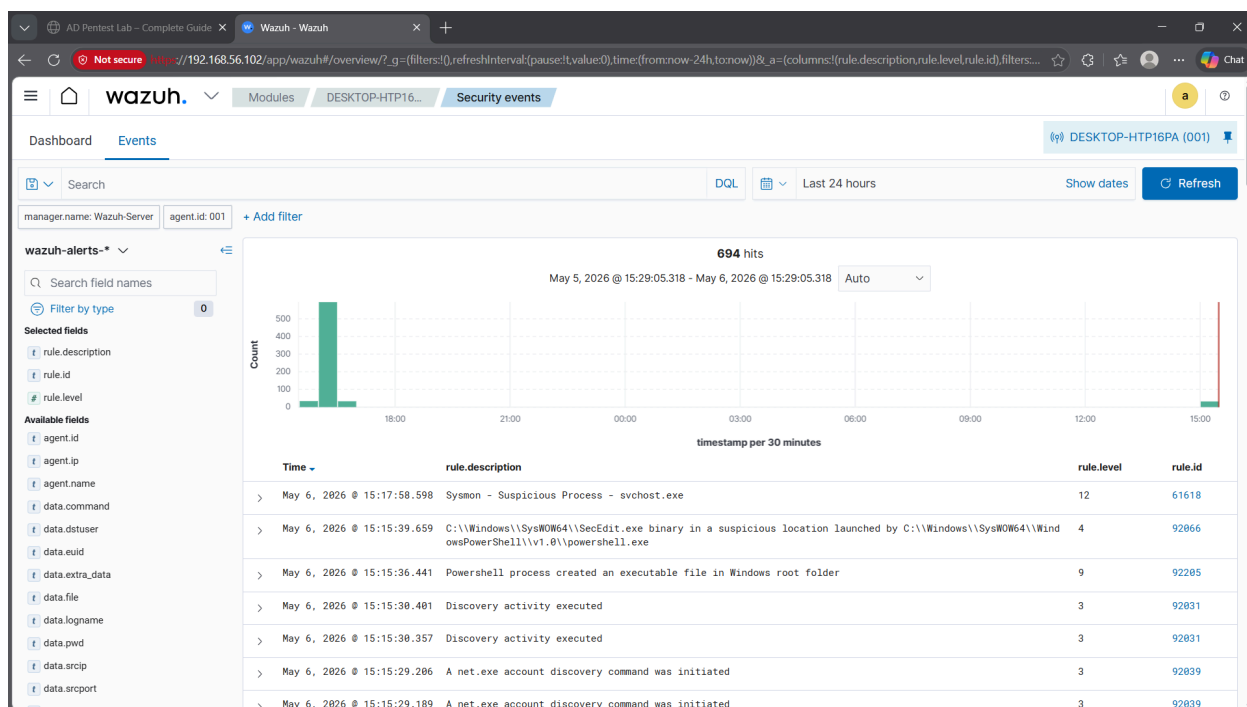


Figure: Wazuh Event Stream of Windows

The Security Events view confirms that Wazuh is receiving alerts from both endpoints. Windows events include logon successes and Sysmon detections, while Ubuntu events include PAM, sudo, and integrity monitoring alerts.

2.3 File Integrity Monitoring (FIM) in Action

FIM watches critical directories and fires an alert any time a file is created, modified, or deleted. This is a foundational detection capability. File Integrity Monitoring (FIM) was configured to monitor Linux system directories such as /etc and /usr/bin. The test modified /etc/hosts by adding a fake entry:

```
192.168.56.1 evil.com
```

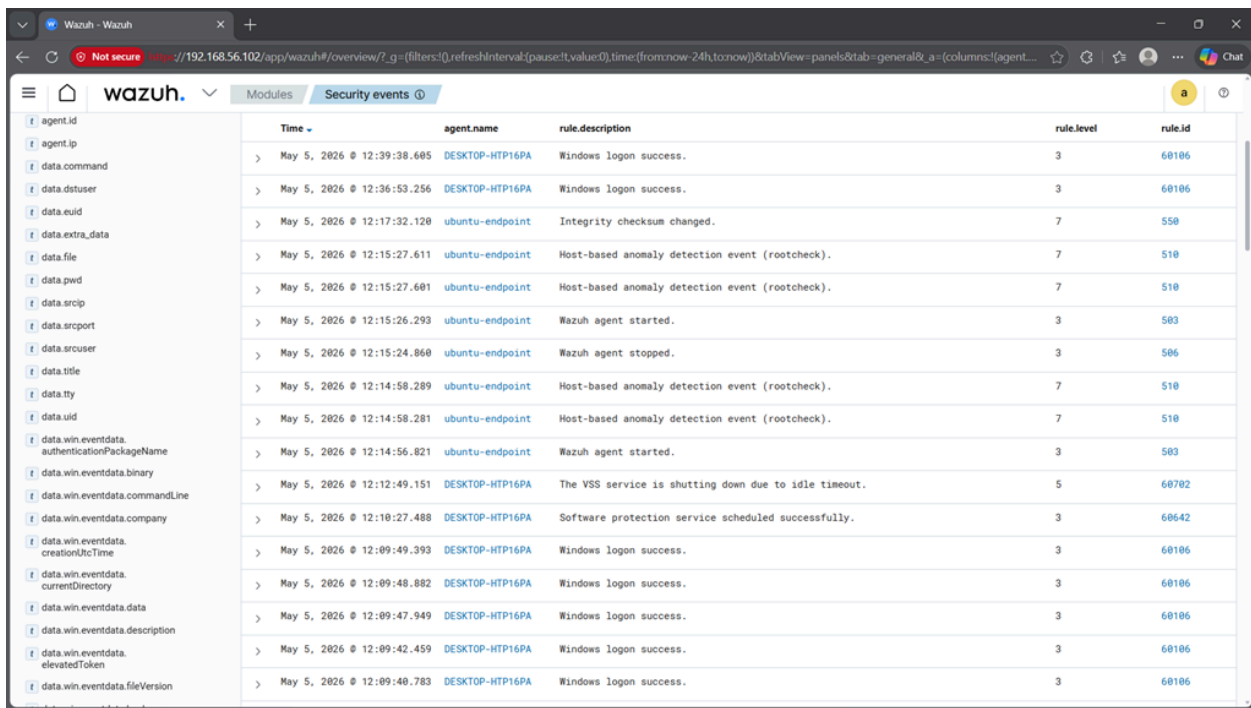


Figure: Wazuh FIM alert showing Integrity checksum changed on ubuntu-endpoint.

Time	agent.name	rule.description	rule.level	rule.id
> May 5, 2026 @ 12:17:32.128	ubuntu-endpoint	Integrity checksum changed.	7	550
> May 5, 2026 @ 12:15:27.611	ubuntu-endpoint	Host-based anomaly detection event (rootcheck).	7	510
> May 5, 2026 @ 12:15:27.681	ubuntu-endpoint	Host-based anomaly detection event (rootcheck).	7	510
> May 5, 2026 @ 12:15:26.293	ubuntu-endpoint	Wazuh agent started.	3	503
> May 5, 2026 @ 12:15:24.868	ubuntu-endpoint	Wazuh agent stopped.	3	506
> May 5, 2026 @ 12:14:58.289	ubuntu-endpoint	Host-based anomaly detection event (rootcheck).	7	510
> May 5, 2026 @ 12:14:58.281	ubuntu-endpoint	Host-based anomaly detection event (rootcheck).	7	510
> May 5, 2026 @ 12:14:56.821	ubuntu-endpoint	Wazuh agent started.	3	503
> May 5, 2026 @ 12:12:49.151	DESKTOP-HTP16PA	The VSS service is shutting down due to idle timeout.	5	60702
> May 5, 2026 @ 12:18:27.488	DESKTOP-HTP16PA	Software protection service scheduled successfully.	3	60642
> May 5, 2026 @ 12:09:49.393	DESKTOP-HTP16PA	Windows login success.	3	60106
> May 5, 2026 @ 12:09:48.882	DESKTOP-HTP16PA	Windows login success.	3	60106
> May 5, 2026 @ 12:09:47.949	DESKTOP-HTP16PA	Windows login success.	3	60106
> May 5, 2026 @ 12:09:42.459	DESKTOP-HTP16PA	Windows login success.	3	60106
> May 5, 2026 @ 12:09:40.783	DESKTOP-HTP16PA	Windows login success.	3	60106
> May 5, 2026 @ 12:09:39.394	DESKTOP-HTP16PA	Windows login success.	3	60106
> May 5, 2026 @ 12:09:39.283	DESKTOP-HTP16PA	Windows login success.	3	60106

Figure: FIM

The FIM alert confirmed that Wazuh detected a file modification on ubuntu-endpoint. The alert generated Rule ID 550, “Integrity checksum changed,” with level 7 severity. The expanded FIM alert shows the affected file path, syscheck event details, reporting agent, and integrity-change evidence. This confirms that Wazuh detected the modification of a monitored Linux system file.

targetFilename	manager.name	rule.description
data.win.eventdata.targetLogonId	Wazuh-Server	Integrity checksum changed.
data.win.eventdata.targetUserSid	# rule.firedtimes	1
data.win.eventdata.targetUserName	rule.gdpr	II.5.1.f
data.win.eventdata.targetUserSid	rule.gpg13	4.11
data.win.eventdata.user	rule.groups	ossec, syscheck, syscheck_entry_modified, syscheck_file
data.win.eventdata.unicodeTime	rule.hipaa	164.312.c.1, 164.312.c.2
data.win.eventdata.virtualAccount	rule.id	550
data.win.system.channel	# rule.level	7
data.win.system.computer	rule.mail	false
data.win.system.eventID	rule.mitre.id	T1565.001
data.win.system.eventRecordID	rule.mitre.tactic	Impact
data.win.system.keywords	rule.mitre.technique	Stored Data Manipulation
data.win.system.level	rule.nist_800_53	SI.7
data.win.system.message	rule.pci_dss	11.5
data.win.system.opcode	rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
data.win.system.processID	syscheck.changed_attributes	size, mtime, md5, sha1, sha256
data.win.system.providerGuid	syscheck.event	modified
data.win.system.providerName		
data.win.system.severityValue		
data.win.system.systemTime		
data.win.system.task		
data.win.system.threadID		
data.win.system.version		

Figure: FIM open alert

SECTION 3 — ALERT INVESTIGATION (SOC TRIAGE PROCESS)

When an alert fires we do not immediately escalate or close it. We perform **triage** — a structured investigation process to determine whether the alert is a true threat. The following documents the full triage of two high-severity alerts detected in this lab.

ALERT 1 — Multiple Failed Login Attempts (Brute Force)

Step 1 — WHAT is the Alert?

An alert is a notification or warning signal generated by a system to inform users or administrators of significant security events requiring investigation.

The alert detected repeated SSH authentication failures caused by invalid usernames and wrong password attempts. This behaviour simulates brute-force or password-guessing activity against a Linux endpoint.

Field	Value
Rule Name	sshd: Attempt to login using a non-existent user / User missed the password more than one time
Rule IDs Observed	5710, 5503, 2502
Severity Level	Level 5 for invalid user and PAM failure; Level 10 for repeated missed password
Alert Time	May 5, 2026 around 09:24–09:25, based on Wazuh screenshot
Rule Category	SSH authentication failure / credential access behavior

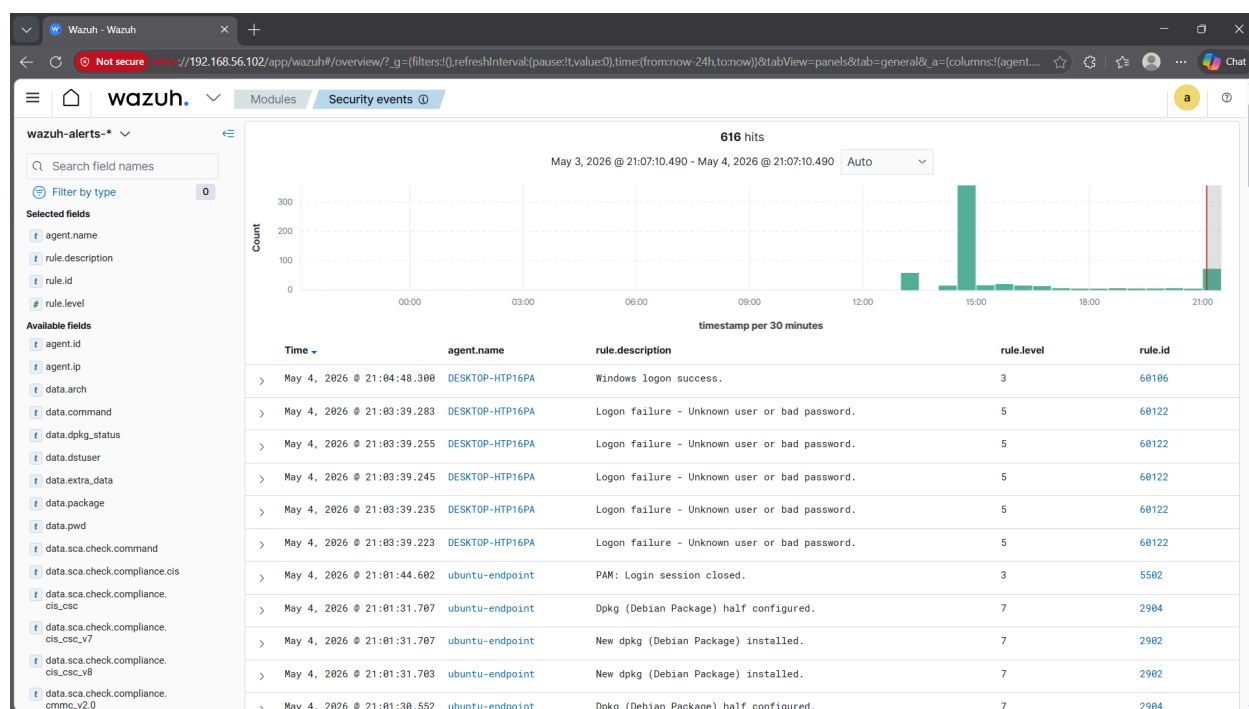


Fig: Brute force/multiple failed login alert

Step 2 — WHO and WHERE?

Field	Value
Agent Name	Wazuh-Server shown in this screenshot; Ubuntu endpoint was also tested separately.
Agent OS	Ubuntu Server 22.04 LTS
Target Username	wronguser / enduser during test attempts
Source IP	Visible in the expanded raw log/auth.log, depending on the alert source.
Destination Port	SSH port 22

The screenshot shows the Wazuh web interface with the MITRE ATT&CK module selected. The main panel displays an alert for a failed login attempt. The alert details are as follows:

Time	agent.name	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id
May 5, 2026 @ 12:50:23.089	ubuntu-endpoint	T1110.001	Credential Access	PAM: User login failed.	5	5583

Below the alert details, there is an "Expanded document" section with a "Table" view. The table contains the following data:

Field	Value
_index	wazuh-alerts-4.x-2026.05.05
agent.id	002
agent.ip	192.168.56.183
agent.name	ubuntu-endpoint
data.dstuser	jazz
data.euid	0
data.logname	jazz
data.srcuser	jazz
data.tty	/dev/pts/0
data.uid	1000
decoder.name	pam
full_log	May 5 07:20:23 ubuntu-endpoint sudo: pam_unix(sudo:auth): authentication failure; logname=jazz uid=1000 euid=0 tty=/dev/pts/0 ruser=jazz rhost= user=jazz
id	1777965623.449371
input.type	log

Fig: Brute force/multiple failed login alert details pt.1

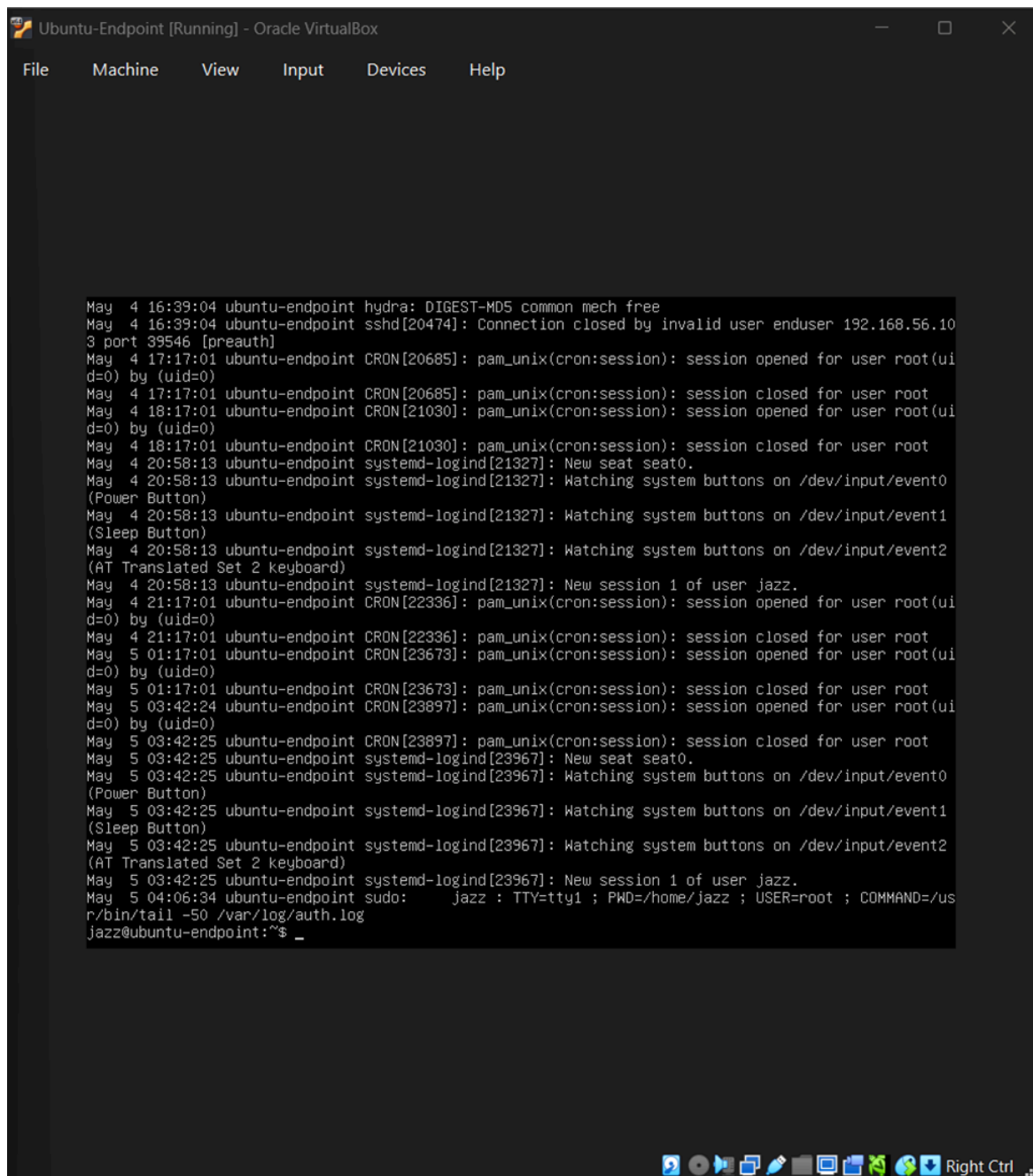
The screenshot shows the Wazuh web interface with the MITRE ATT&CK module selected. The main panel displays an alert for a failed login attempt. The alert details are as follows:

Field	Value
predecoder.program_name	sudo
predecoder.timestamp	May 5 07:20:23
rule.description	PAM: User login failed.
rule.firedtimes	1
rule.gdpr	IV_35.7.d, IV_32.2
rule.gpg13	7,8
rule.groups	pam, syslog, authentication_failed
rule.hipaa	164.312.b
rule.id	5583
rule.level	5
rule.mail	false
rule.mitre.id	T1110.001
rule.mitre.tactic	Credential Access
rule.mitre.technique	Password Guessing
rule.nist_800_53	AU.14, AC.7
rule.pci_dss	10.2.4, 10.2.5
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
timestamp	May 5, 2026 @ 12:50:23.089

Fig: Brute force/multiple failed login alert pt.2

Step 3 — WHAT Happened Exactly?

SSH login attempts were generated using invalid usernames and wrong passwords. This simulates a basic brute-force or password-guessing attempt. The Ubuntu auth log confirmed invalid user activity, and Wazuh generated SSH/PAM authentication alerts.



```
May  4 16:39:04 ubuntu-endpoint hydra: DIGEST-MD5 common mech free
May  4 16:39:04 ubuntu-endpoint sshd[20474]: Connection closed by invalid user enduser 192.168.56.10
3 port 39546 [preauth]
May  4 17:17:01 ubuntu-endpoint CRON[20685]: pam_unix(cron:session): session opened for user root(ui
d=0) by (uid=0)
May  4 17:17:01 ubuntu-endpoint CRON[20685]: pam_unix(cron:session): session closed for user root
May  4 18:17:01 ubuntu-endpoint CRON[21030]: pam_unix(cron:session): session opened for user root(ui
d=0) by (uid=0)
May  4 18:17:01 ubuntu-endpoint CRON[21030]: pam_unix(cron:session): session closed for user root
May  4 20:58:13 ubuntu-endpoint systemd-logind[21327]: New seat seat0.
May  4 20:58:13 ubuntu-endpoint systemd-logind[21327]: Watching system buttons on /dev/input/event0
(Power Button)
May  4 20:58:13 ubuntu-endpoint systemd-logind[21327]: Watching system buttons on /dev/input/event1
(Sleep Button)
May  4 20:58:13 ubuntu-endpoint systemd-logind[21327]: Watching system buttons on /dev/input/event2
(AT Translated Set 2 keyboard)
May  4 20:58:13 ubuntu-endpoint systemd-logind[21327]: New session 1 of user jazz.
May  4 21:17:01 ubuntu-endpoint CRON[22336]: pam_unix(cron:session): session opened for user root(ui
d=0) by (uid=0)
May  4 21:17:01 ubuntu-endpoint CRON[22336]: pam_unix(cron:session): session closed for user root
May  5 01:17:01 ubuntu-endpoint CRON[23673]: pam_unix(cron:session): session opened for user root(ui
d=0) by (uid=0)
May  5 01:17:01 ubuntu-endpoint CRON[23673]: pam_unix(cron:session): session closed for user root
May  5 03:42:24 ubuntu-endpoint CRON[23897]: pam_unix(cron:session): session opened for user root(ui
d=0) by (uid=0)
May  5 03:42:25 ubuntu-endpoint CRON[23897]: pam_unix(cron:session): session closed for user root
May  5 03:42:25 ubuntu-endpoint systemd-logind[23967]: New seat seat0.
May  5 03:42:25 ubuntu-endpoint systemd-logind[23967]: Watching system buttons on /dev/input/event0
(Power Button)
May  5 03:42:25 ubuntu-endpoint systemd-logind[23967]: Watching system buttons on /dev/input/event1
(Sleep Button)
May  5 03:42:25 ubuntu-endpoint systemd-logind[23967]: Watching system buttons on /dev/input/event2
(AT Translated Set 2 keyboard)
May  5 03:42:25 ubuntu-endpoint systemd-logind[23967]: New session 1 of user jazz.
May  5 04:06:34 ubuntu-endpoint sudo:      jazz : TTY=ttty1 ; PWD=/home/jazz ; USER=root ; COMMAND=/us
r/bin/tail -50 /var/log/auth.log
jazz@ubuntu-endpoint:~$ _
```

Figure: Ubuntu auth.log evidence showing SSH invalid user / authentication-related logs.

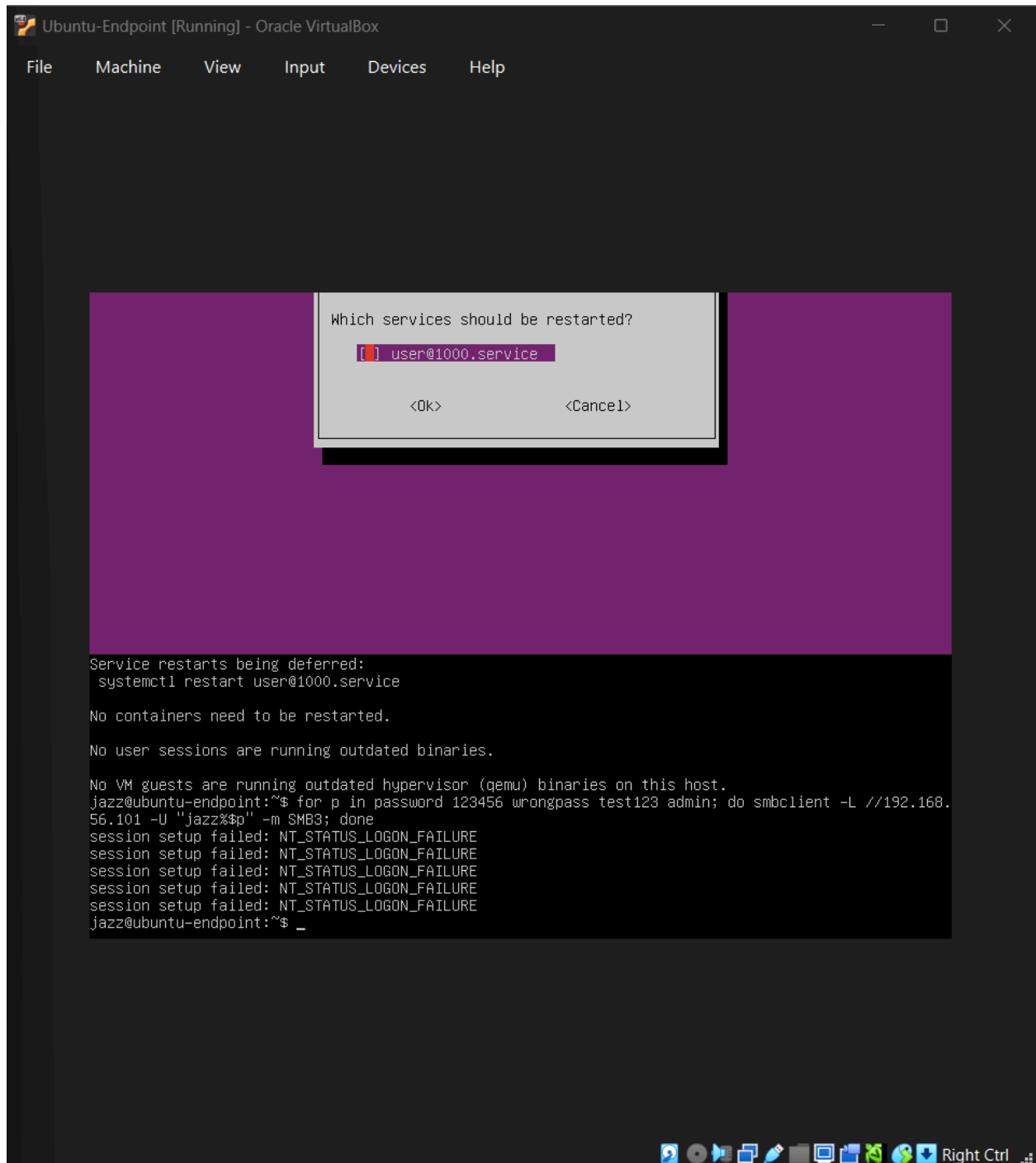
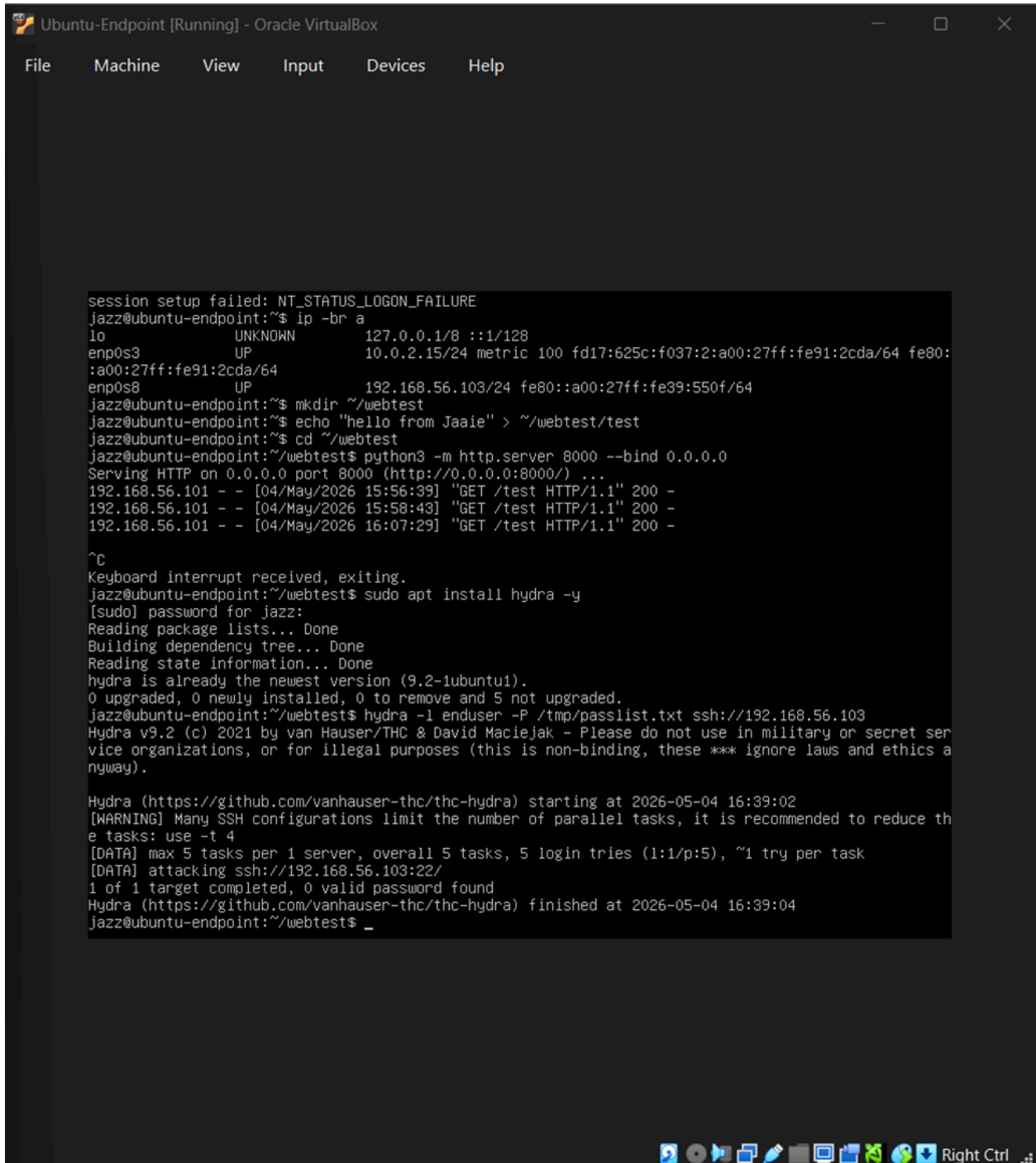


Figure: Failed SSH login attempts shown in Ubuntu terminal



```
session setup failed: NT_STATUS_LOGON_FAILURE
jazz@ubuntu-endpoint:~$ ip -br a
lo                UNKNOWN    127.0.0.1/8 ::1/128
enp0s3            UP          10.0.2.15/24 metric 100 fd17:625c:f037:2:a00:27ff:fe91:2cda/64 fe80::a00:27ff:fe91:2cda/64
enp0s8            UP          192.168.56.103/24 fe80::a00:27ff:fe39:550f/64
jazz@ubuntu-endpoint:~$ mkdir ~/webtest
jazz@ubuntu-endpoint:~$ echo "hello from Jaaie" > ~/webtest/test
jazz@ubuntu-endpoint:~$ cd ~/webtest
jazz@ubuntu-endpoint:~/webtest$ python3 -m http.server 8000 --bind 0.0.0.0
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.56.101 - - [04/May/2026 15:56:39] "GET /test HTTP/1.1" 200 -
192.168.56.101 - - [04/May/2026 15:58:43] "GET /test HTTP/1.1" 200 -
192.168.56.101 - - [04/May/2026 16:07:29] "GET /test HTTP/1.1" 200 -
^C
Keyboard interrupt received, exiting.
jazz@ubuntu-endpoint:~/webtest$ sudo apt install hydra -y
[sudo] password for jazz:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
hydra is already the newest version (9.2-1ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
jazz@ubuntu-endpoint:~/webtest$ hydra -l enduser -P /tmp/passlist.txt ssh://192.168.56.103
Hydra v9.2 (c) 2021 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-04 16:39:02
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 5 tasks per 1 server, overall 5 tasks, 5 login tries (1:1/p:5), ~1 try per task
[DATA] attacking ssh://192.168.56.103:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-04 16:39:04
jazz@ubuntu-endpoint:~/webtest$ _
```

Figure: Hydra SSH test showing password attempts against the Ubuntu endpoint.

Step 4 — MITRE ATT&CK Mapping

Field	Value
Technique	T1110.001 — Password Guessing
Tactic	Credential Access
Attacker Next Step	If credentials succeed, attacker moves to Initial Access, command execution, persistence, and privilege escalation.

The screenshot displays the Wazuh web interface, specifically the MITRE ATT&CK mapping section. The interface shows a rule for a PAM login failure event. The rule details are as follows:

Time	agent.name	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id
May 5, 2026 @ 12:50:23.089	ubuntu-endpoint	T1110.001	Credential Access	PAM: User login failed.	5	5503

Below the rule details, there is an "Expanded document" section showing a JSON representation of the event data. The JSON data includes fields such as _index, agent.id, agent.ip, agent.name, data.dstuser, data.euid, data.logname, data.srcuser, data.tty, data.uid, data.win.eventdata.authenticationPackageName, data.win.eventdata.commandLine, data.win.eventdata.company, data.win.eventdata.creationUtcTime, data.win.eventdata.currentDirectory, data.win.eventdata.data, data.win.eventdata.description, data.win.eventdata.elevatedToken, and data.win.eventdata.fileVersion.

Figure: MITRE ATT&CK mapping for brute force — part 1

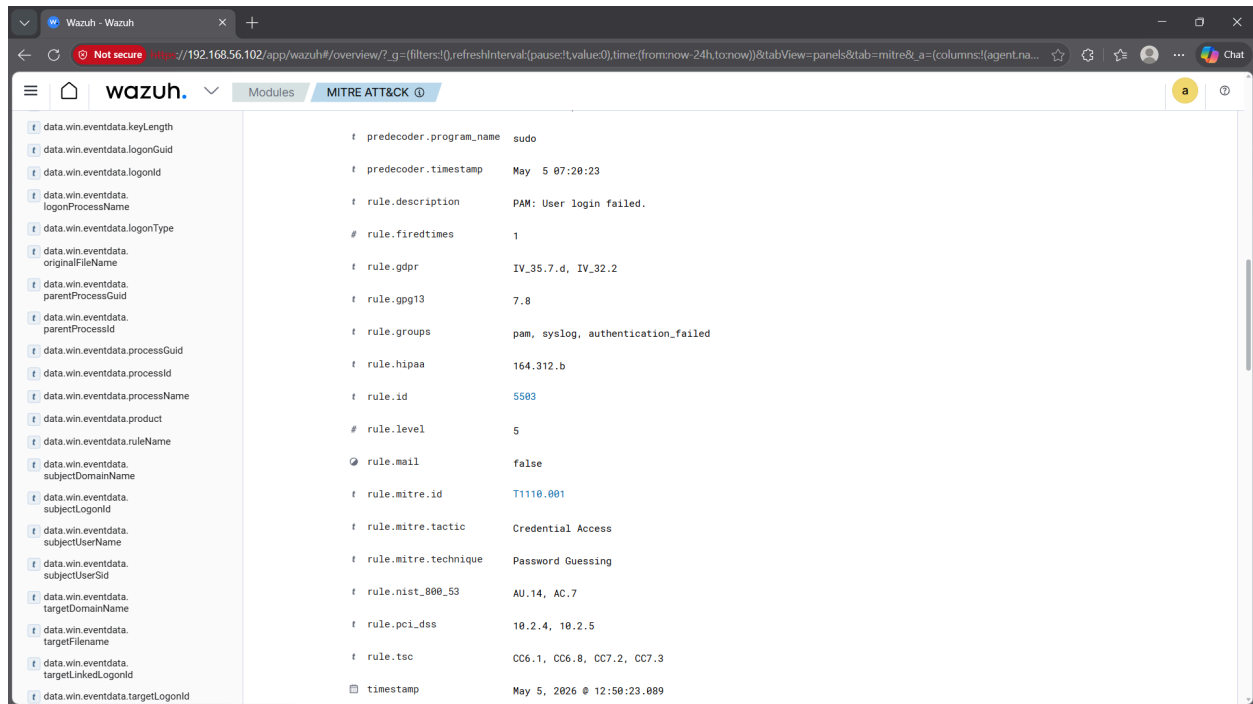


Figure: MITRE ATT&CK mapping for brute force — part 2

Step 5 — VERDICT

Verdict: TRUE POSITIVE — Lab Simulation. The login attempts were intentionally generated, and Wazuh correctly detected authentication failures. In a real SOC, repeated invalid user attempts should be treated as suspicious, especially if coming from an unknown source IP or occurring rapidly.

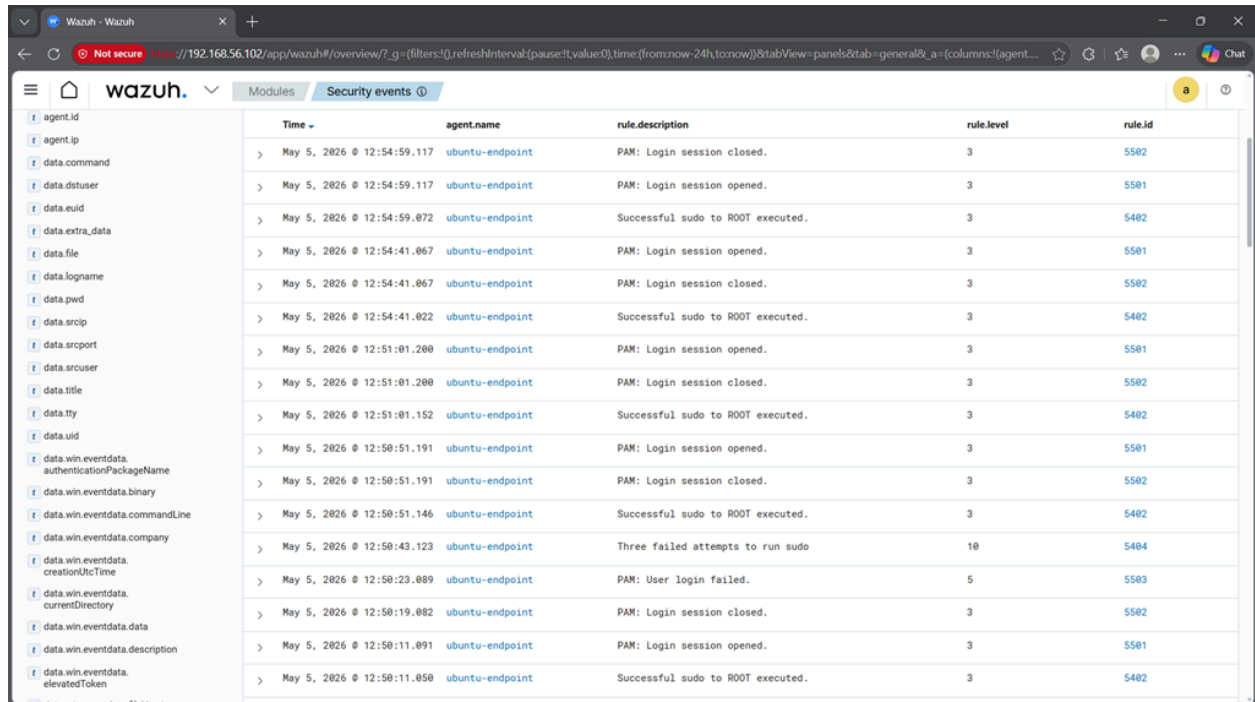
Step 6 — RESPONSE ACTIONS

1. Check for successful logins from the same source after failed attempts.
2. Block the source IP if malicious and repeated.
3. Enable account lockout or fail2ban for SSH-facing systems.
4. Disable password-based SSH where possible and enforce SSH keys.
5. Review all Linux agents for the same source IP to detect lateral movement.

ALERT 2 — Privilege Escalation / Suspicious sudo Usage (Linux)

Step 1 — WHAT is the Alert?

Field	Value
Rule Name	Successful sudo to ROOT executed / Three failed attempts to run sudo
Rule IDs	5402 and 5404
Severity Level	Rule 5402 Level 3; Rule 5404 Level 10
Alert Time	May 5, 2026 around 12:50–12:54, based on Wazuh Security Events screenshot
Rule Category	Linux sudo monitoring / privilege escalation behavior



	Time	agent.name	rule.description	rule.level	rule.id
	> May 5, 2026 @ 12:54:59.117	ubuntu-endpoint	PAM: Login session closed.	3	5502
	> May 5, 2026 @ 12:54:59.117	ubuntu-endpoint	PAM: Login session opened.	3	5501
	> May 5, 2026 @ 12:54:59.072	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5402
	> May 5, 2026 @ 12:54:41.067	ubuntu-endpoint	PAM: Login session opened.	3	5501
	> May 5, 2026 @ 12:54:41.067	ubuntu-endpoint	PAM: Login session closed.	3	5502
	> May 5, 2026 @ 12:54:41.022	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5402
	> May 5, 2026 @ 12:51:01.200	ubuntu-endpoint	PAM: Login session opened.	3	5501
	> May 5, 2026 @ 12:51:01.200	ubuntu-endpoint	PAM: Login session closed.	3	5502
	> May 5, 2026 @ 12:51:01.152	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5402
	> May 5, 2026 @ 12:50:51.191	ubuntu-endpoint	PAM: Login session opened.	3	5501
	> May 5, 2026 @ 12:50:51.191	ubuntu-endpoint	PAM: Login session closed.	3	5502
	> May 5, 2026 @ 12:50:51.146	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5402
	> May 5, 2026 @ 12:50:43.123	ubuntu-endpoint	Three failed attempts to run sudo	10	5404
	> May 5, 2026 @ 12:50:23.089	ubuntu-endpoint	PAM: User login failed.	5	5503
	> May 5, 2026 @ 12:50:19.082	ubuntu-endpoint	PAM: Login session closed.	3	5502
	> May 5, 2026 @ 12:50:11.091	ubuntu-endpoint	PAM: Login session opened.	3	5501
	> May 5, 2026 @ 12:50:11.050	ubuntu-endpoint	Successful sudo to ROOT executed.	3	5402

Figure: Wazuh Security Events showing successful sudo and failed sudo attempts on ubuntu-endpoint.

Step 2 — WHO and WHERE?

Field	Value
Agent Name	ubuntu-endpoint
Agent OS	Ubuntu 22.04.5 LTS
User Account	jazz
Target Account	root
Commands Observed	sudo ls /root, sudo cat /etc/shadow, sudo su root

Step 3 — WHAT Happened Exactly?

The user jazz executed sudo commands to access root-only resources, including /root and /etc/shadow. The test also simulated failed sudo authentication by entering an incorrect password three times. Wazuh detected both successful privilege elevation and repeated failed sudo attempts.

```
jazz@ubuntu-endpoint: ~  
for user root(uid=0) by jazz(uid=1000)  
May  5 06:52:02 ubuntu-endpoint sudo: pam_unix(sudo:session): session closed  
for user root  
May  5 06:55:19 ubuntu-endpoint sudo:    jazz : TTY=pts/0 ; PWD=/home/jazz  
; USER=root ; COMMAND=/usr/bin/cat /etc/shadow  
May  5 06:55:19 ubuntu-endpoint sudo: pam_unix(sudo:session): session opened  
for user root(uid=0) by jazz(uid=1000)  
May  5 06:55:19 ubuntu-endpoint sudo: pam_unix(sudo:session): session closed  
for user root  
May  5 06:55:38 ubuntu-endpoint sudo:    jazz : TTY=pts/0 ; PWD=/home/jazz  
; USER=root ; COMMAND=/usr/bin/tail /var/log/auth.log  
May  5 06:55:38 ubuntu-endpoint sudo: pam_unix(sudo:session): session opened  
for user root(uid=0) by jazz(uid=1000)  
May  5 06:55:38 ubuntu-endpoint sudo: pam_unix(sudo:session): session closed  
for user root  
May  5 06:55:50 ubuntu-endpoint sudo:    jazz : TTY=pts/0 ; PWD=/home/jazz  
; USER=root ; COMMAND=/usr/bin/su root  
May  5 06:55:50 ubuntu-endpoint sudo: pam_unix(sudo:session): session opened  
for user root(uid=0) by jazz(uid=1000)  
May  5 07:05:22 ubuntu-endpoint sudo: pam_unix(sudo:session): session closed  
for user root  
May  5 07:05:43 ubuntu-endpoint sudo:    jazz : TTY=pts/0 ; PWD=/home/jazz  
; USER=root ; COMMAND=/usr/bin/ls /root  
May  5 07:05:43 ubuntu-endpoint sudo: pam_unix(sudo:session): session opened  
for user root(uid=0) by jazz(uid=1000)  
May  5 07:05:43 ubuntu-endpoint sudo: pam_unix(sudo:session): session closed  
for user root  
May  5 07:06:51 ubuntu-endpoint sudo:    jazz : TTY=pts/0 ; PWD=/home/jazz  
; USER=root ; COMMAND=/usr/bin/ls /root  
May  5 07:06:51 ubuntu-endpoint sudo: pam_unix(sudo:session): session opened  
for user root(uid=0) by jazz(uid=1000)  
May  5 07:06:51 ubuntu-endpoint sudo: pam_unix(sudo:session): session closed  
for user root  
May  5 07:07:25 ubuntu-endpoint sudo: pam_unix(sudo:auth): authentication fa  
ilure; logname=jazz uid=1000 euid=0 tty=/dev/pts/0 ruser=jazz rhost= user=j  
azz  
May  5 07:07:37 ubuntu-endpoint sudo:    jazz : 3 incorrect password attemp  
ts ; TTY=pts/0 ; PWD=/home/jazz ; USER=root ; COMMAND=/usr/bin/ls /root  
May  5 07:07:54 ubuntu-endpoint sudo:    jazz : TTY=pts/0 ; PWD=/home/jazz  
; USER=root ; COMMAND=/usr/bin/tail -50 /var/log/auth.log  
May  5 07:07:54 ubuntu-endpoint sudo: pam_unix(sudo:session): session opened  
for user root(uid=0) by jazz(uid=1000)  
jazz@ubuntu-endpoint:~$
```

Figure: Ubuntu terminal/auth.log evidence showing successful sudo commands and three incorrect sudo password attempts

Step 4 — MITRE ATT&CK Mapping

Field	Value
Technique	T1548.003 — Sudo and Sudo Caching
Tactic	Privilege Escalation / Defense Evasion
Attacker Next Step	Modify system files, create users, read credential files, install persistence, disable logging, or run tools as root.

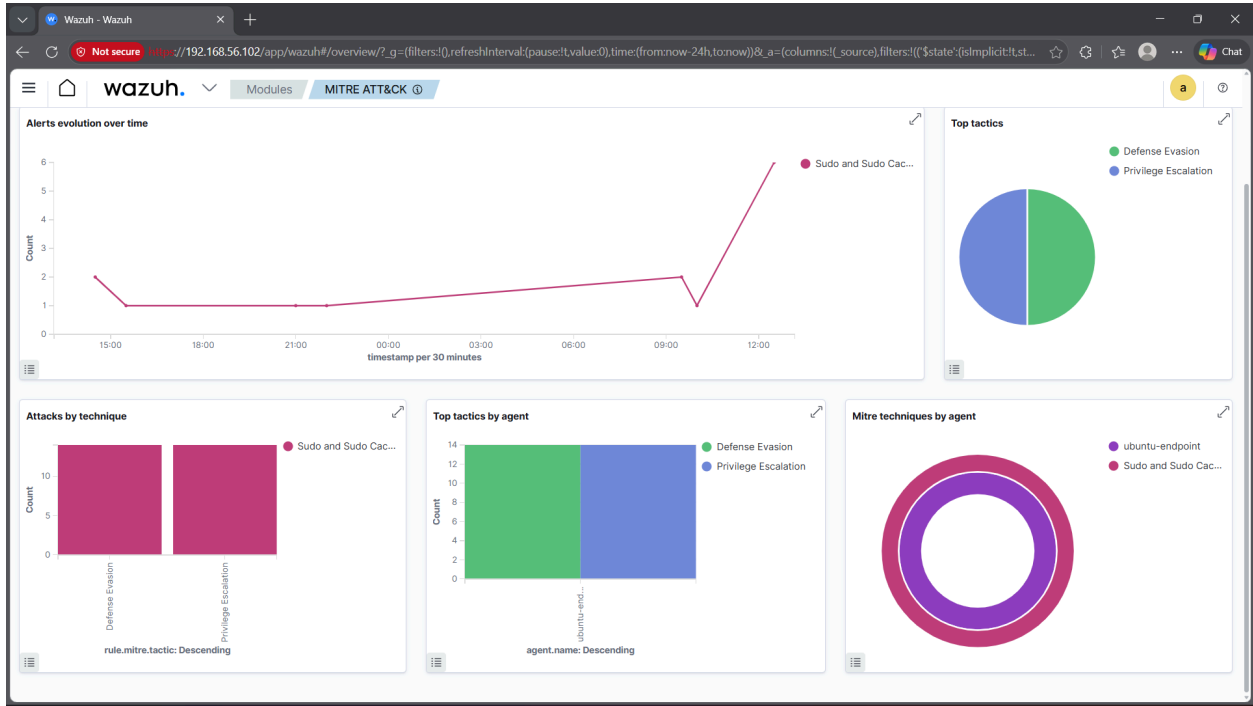


Figure: MITRE ATT&CK Sudo and Sudo Caching

Step 5 — VERDICT

Verdict: TRUE POSITIVE — Lab Simulation. The sudo activity occurred exactly as tested. It is benign in the lab, but in production this would require context review. Successful sudo can be normal for administrators, but failed sudo attempts combined with access to /etc/shadow significantly increase the risk.

Step 6 — RESPONSE ACTIONS

1. Review `/var/log/auth.log` for the full sudo session history.
2. Verify whether the user is authorized in `/etc/sudoers` or sudo group membership.
3. Check command history for follow-up root-level changes.
4. If unauthorized, lock the user account and isolate the endpoint.
5. Review `/etc/passwd`, `/etc/shadow`, cron jobs, and new services for persistence.

SECTION 4 — CUSTOM DETECTION RULES

4.1 Why Write Custom Rules?

Wazuh ships with 3,000+ built-in rules. Custom rules extend detection for your specific environment, reduce false positives, and demonstrate rule-writing skill to employers. Rules are written in XML and stored in `/var/ossec/etc/rules/local_rules.xml`.

The following built-in rules were confirmed:

Rule ID	Description	Detection Area	Level
92057	PowerShell encoded command	Windows suspicious command execution	Level 12
92027	PowerShell process spawned PowerShell instance	Windows process behavior	Level 4
92213	Executable file dropped in folder commonly used by malware	Windows file/process behavior	Level 15
5710	sshd: Attempt to login using a non-existent user	Linux SSH authentication	Level 5
2502	User missed the password more than one time	Linux authentication failures	Level 10
5402	Successful sudo to ROOT executed	Linux sudo monitoring	Level 3
5404	Three failed attempts to run sudo	Linux sudo failed attempts	Level 10
550	Integrity checksum changed	FIM / syscheck	Level 7

4.2 Optional Custom Rule Plan

Custom rules were not required to prove the main detections in this lab. For future improvement, the following local_rules.xml entries could be added and tested:

```
<group name="local,linux,reverse_shell,">
  <rule id="100001" level="12">
    <if_group>syslog</if_group>
    <match>nc -e|nc -lvp|/bin/bash -i</match>
    <description>Possible reverse shell - netcat or interactive shell pattern</description>
    <mitre>
    <id>T1059</id>
    </mitre>
  </rule>
</group>
```

```
<group name="local,windows,powershell,">
  <rule id="100002" level="12">
    <if_group>windows,powershell,</if_group>
    <match>-EncodedCommand|-enc</match>
    <description>PowerShell encoded command execution detected</description>
    <mitre>
    <id>T1059.001</id>
    </mitre>
  </rule>
</group>
```

Status: planned improvement. Do not claim these custom rules fired unless you actually add them, restart Wazuh Manager, trigger them, and capture Wazuh alerts with Rule IDs 100001 or 100002.

SECTION 5 — MITRE ATT&CK COVERAGE

5.1 Wazuh MITRE ATT&CK Dashboard

Wazuh includes a built-in MITRE ATT&CK module that visualises which tactics and techniques have been detected in your environment. This is the same view used by enterprise SOC teams to measure detection coverage.

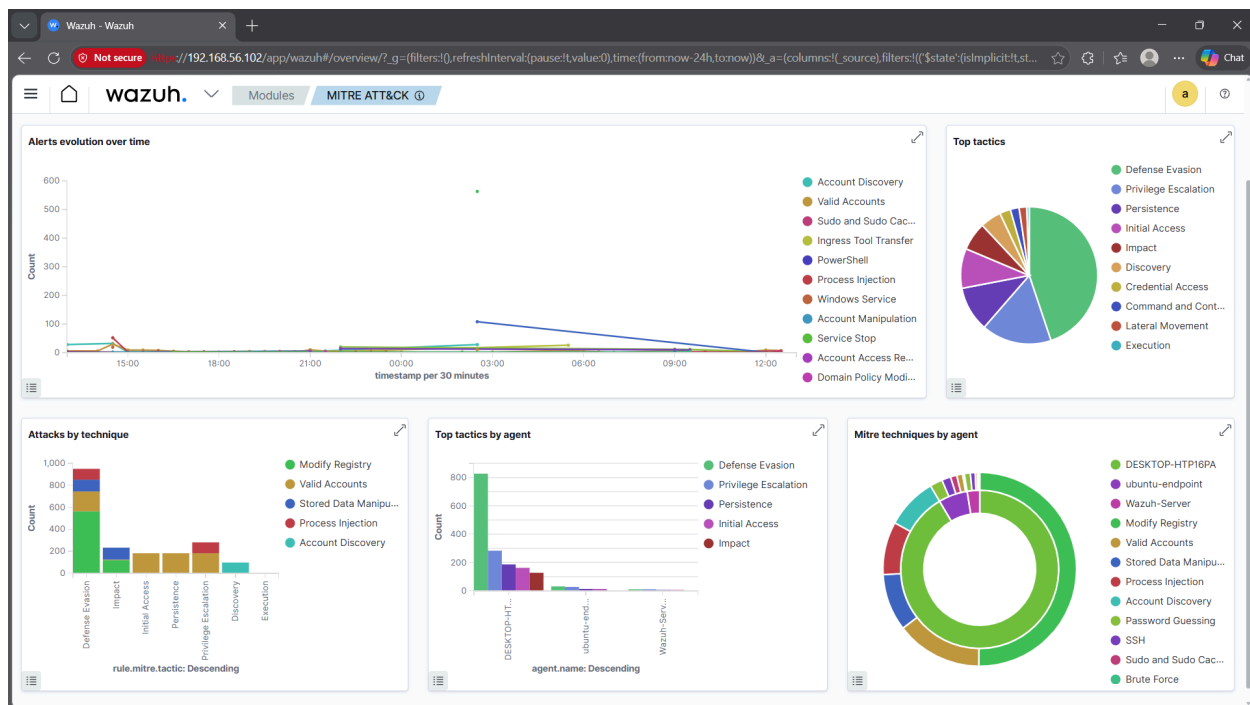


Figure: Wazuh MITRE ATT&CK dashboard showing detected tactics and techniques.

The MITRE ATT&CK dashboard confirms that Wazuh mapped alerts to tactics and techniques such as Defense Evasion, Privilege Escalation, Credential Access, Initial Access, Stored Data Manipulation, PowerShell, SSH, and Password Guessing. This view is useful for showing detection coverage across the lab.

5.2 Techniques Detected in This Lab

ATT&CK ID	Technique	Tactic	Observed Wazuh Rule IDs
T1110.001	Password Guessing / Brute Force	Credential Access	5710, 2502, SSH/PAM alerts
T1078	Valid Accounts	Initial Access	5501, 5503, 60106 depending on event
T1548.003	Sudo and Sudo Caching	Privilege Escalation	5402, 5404
T1059.001	PowerShell	Execution / Defense Evasion	92057, 92027
T1565.001	Stored Data Manipulation	Impact	550
T1055	Process Injection	Defense Evasion / Privilege Escalation	61618
T1570	Lateral Tool Transfer	Lateral Movement	92217

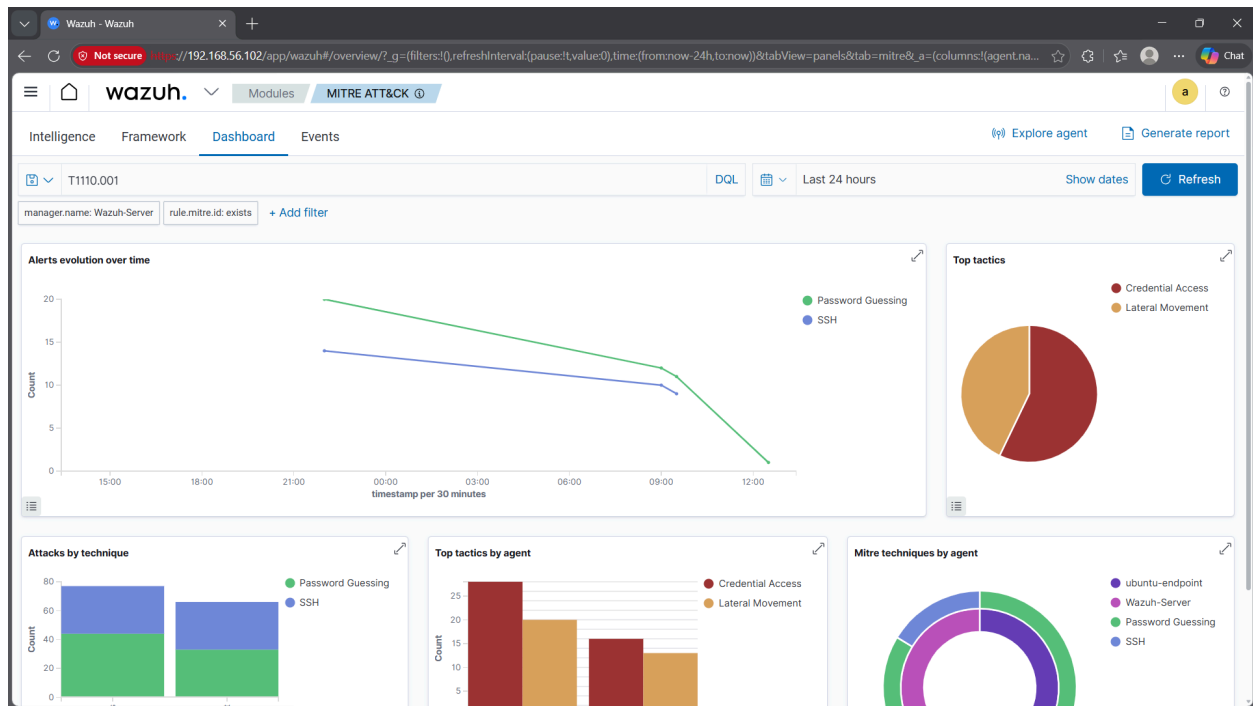


Figure: MITRE ATT&CK Attack Sequence Timeline

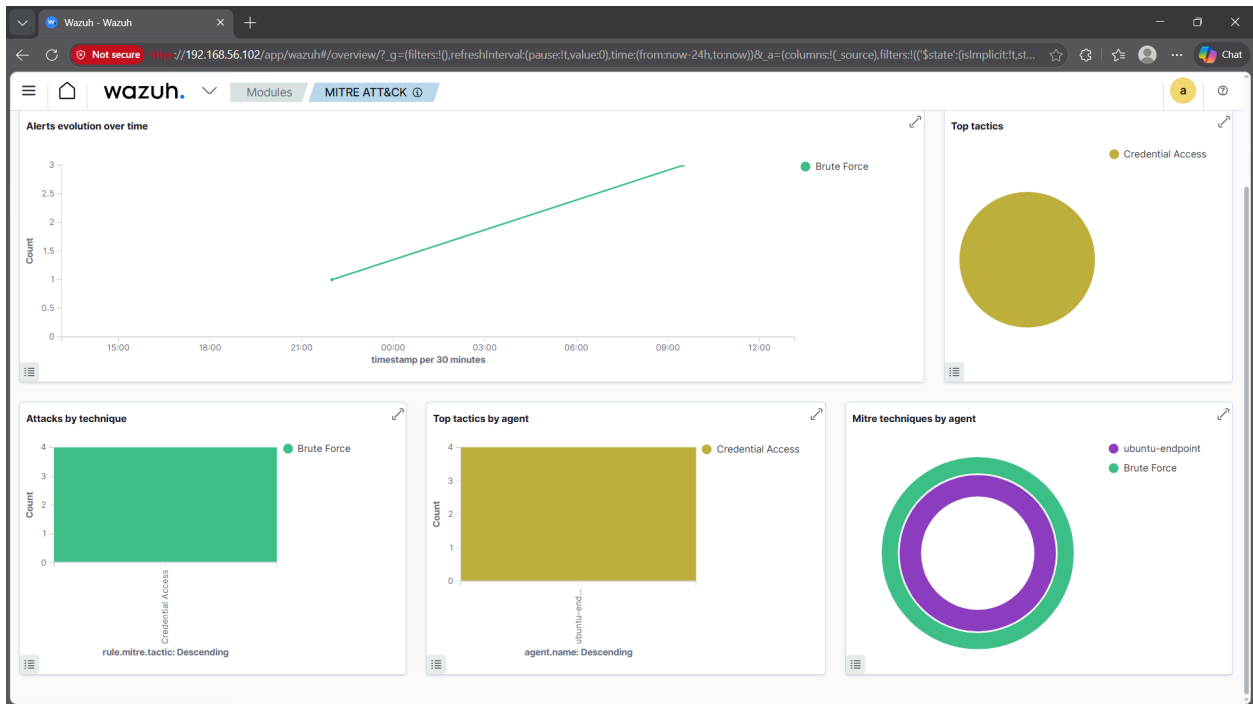


Figure: MITRE ATT&CK Brute Force Attack

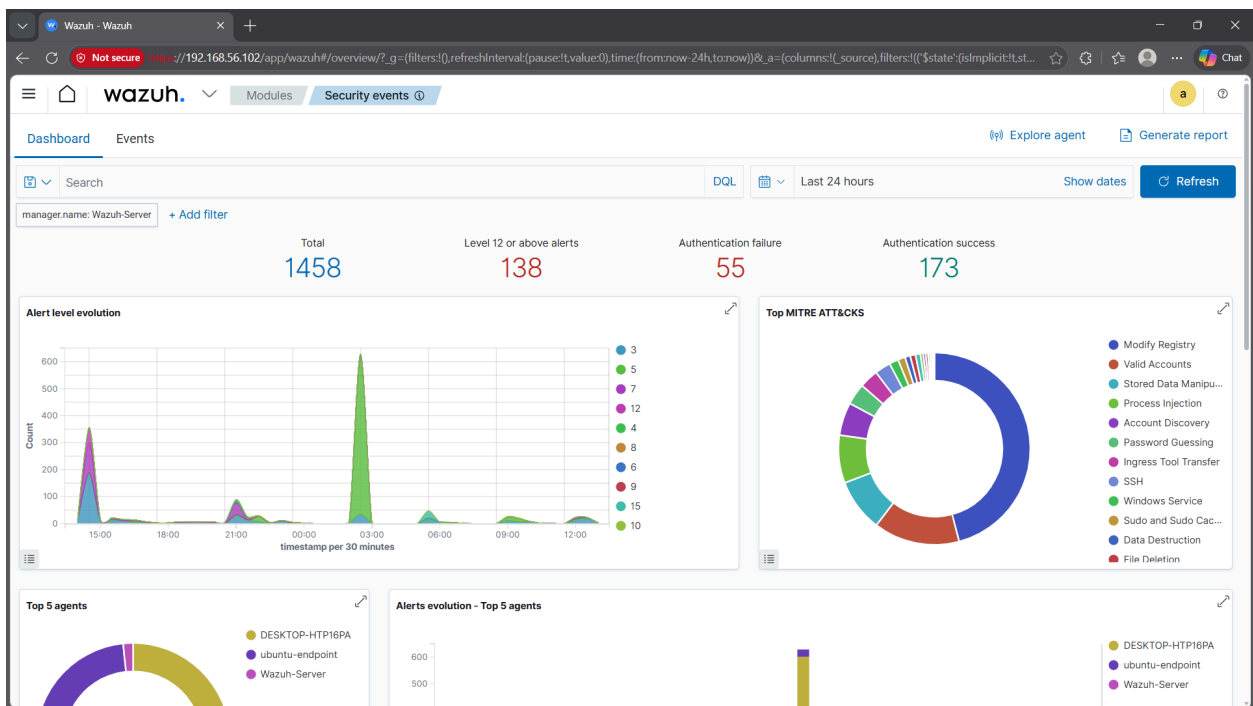


Figure: MITRE ATT&CK Security Event Timeline

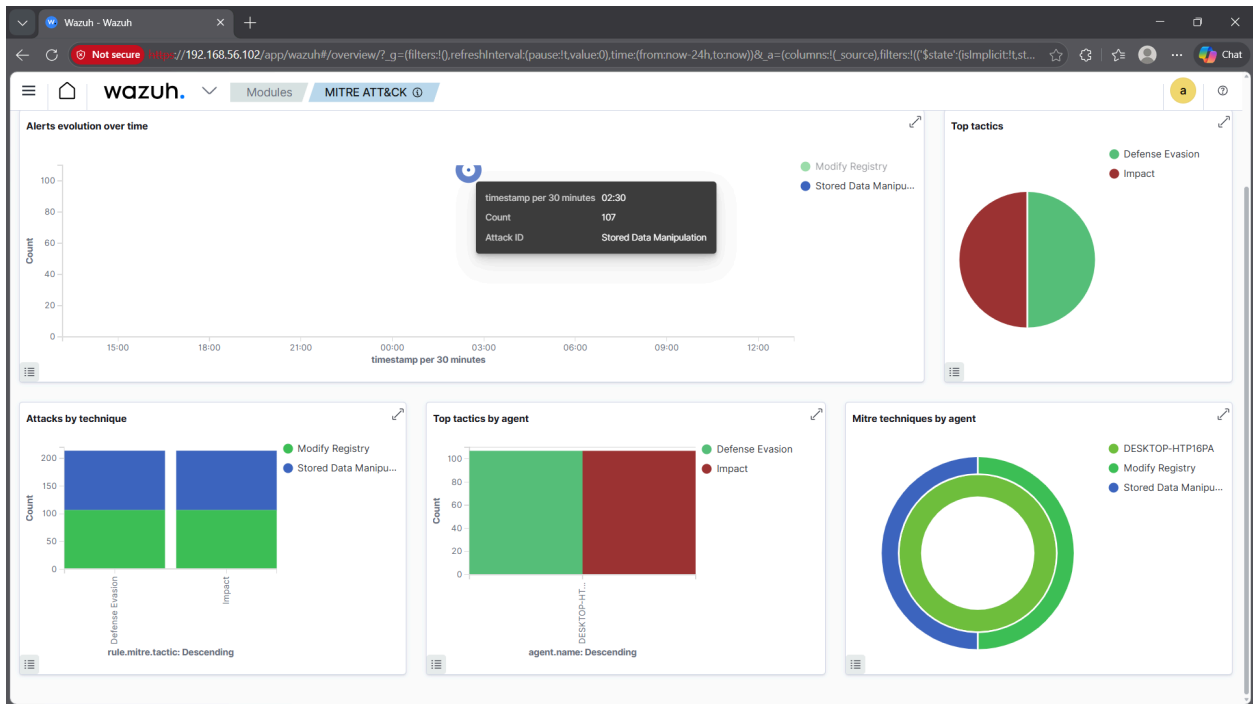


Figure: MITRE ATT&CK Stored Data Manipulation

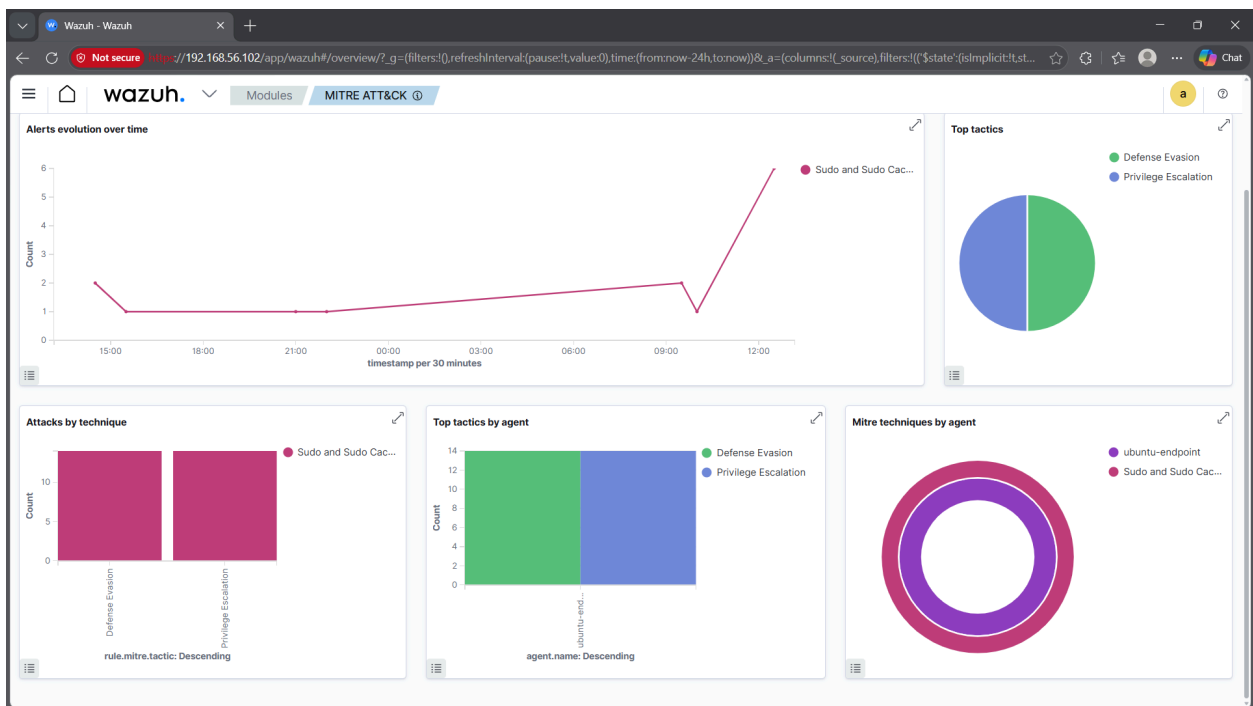


Figure: MITRE ATT&CK Sudo and Sudo Caching

SECTION 6 — WAZUH VULNERABILITY DETECTION

6.1 Vulnerability Scan Result

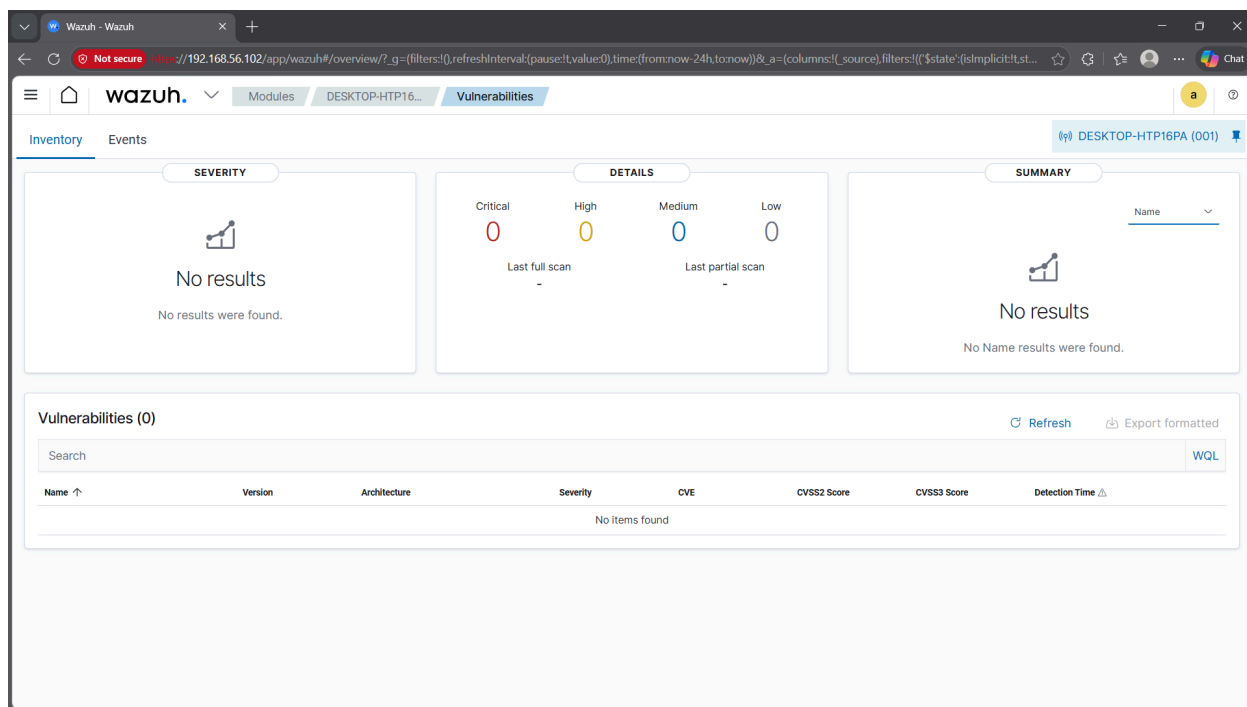


Figure: Initial Wazuh Vulnerability Detection Dashboard before feed update completed

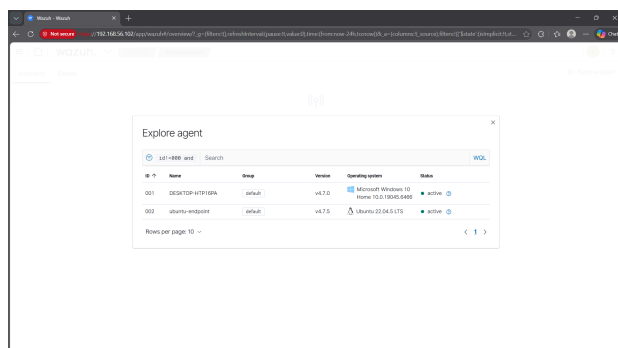


Figure: Vulnerability agent selection view

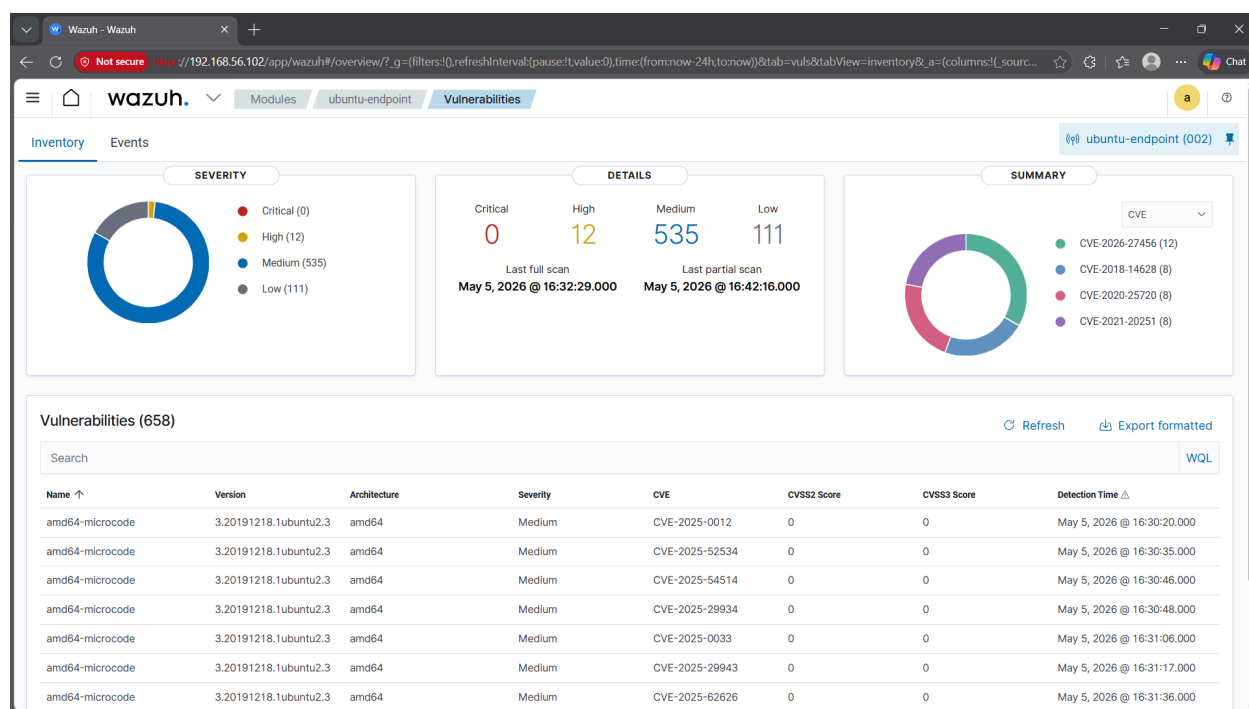


Figure: Wazuh Vulnerability Detection inventory for ubuntu-endpoint.

Field	Value
Agent	ubuntu-endpoint
Operating System	Ubuntu 22.04.5 LTS
Total Vulnerabilities	658
Critical	0
High	12
Medium	535
Low	111
Last Full Scan	May 5, 2026 @ 16:32:29
Last Partial Scan	May 5, 2026 @ 16:42:16

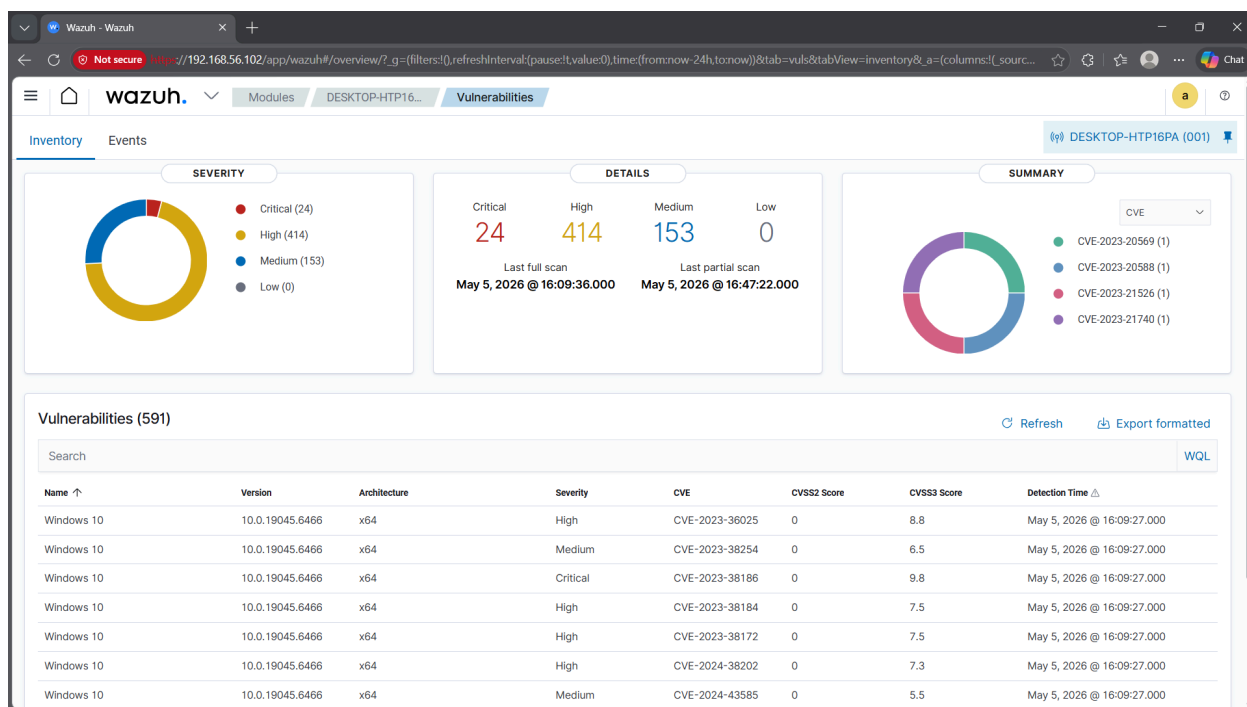


Figure: Wazuh Vulnerability Detection inventory for Windows endpoint

Field	Value
Agent	DESKTOP-HTP16PA
Operating System	Windows 10 Home
Total Vulnerabilities	591
Critical	24
High	414
Medium	153
Low	0
Last Full Scan	May 5, 2026 @ 16:09:36
Last Partial Scan	May 5, 2026 @ 16:47:22

SECTION 7 — LESSONS LEARNED AND ANALYST REFLECTIONS

7.1 Technical Skills Demonstrated

Skill Area	Evidence From Project
SIEM/XDR Deployment	Deployed Wazuh 4.7.5 all-in-one server and connected Windows + Ubuntu agents.
Virtual Networking	Configured host-only networking so VMs communicated on 192.168.56.0/24.
Windows Telemetry	Installed and configured Sysmon and Wazuh agent to detect suspicious PowerShell activity.
Linux Telemetry	Collected auth.log, SSH, PAM, and sudo events from Ubuntu.
FIM	Configured Syscheck/FIM to detect modifications to /etc/hosts.
Vulnerability Detection	Enabled vulnerability scan and confirmed 658 vulnerabilities on Ubuntu endpoint.
SOC Triage	Followed a structured triage process: what, who/where, what happened, MITRE, verdict, response.
MITRE ATT&CK	Mapped alerts to tactics such as Credential Access, Privilege Escalation, Execution, Impact, and Defense Evasion.

7.2 What I Would Improve Next

1. Add a dedicated Kali attacker VM instead of using Ubuntu-Endpoint for some attack tests.
2. Update the Windows Wazuh agent from v4.7.0 to v4.8.0 for better version consistency.
3. Add custom detection rules in local_rules.xml and capture proof of custom rule firing.
4. Enable Wazuh Active Response for automated IP blocking in brute-force scenarios.
5. Add email/Slack alerting for level 12+ alerts to simulate SOC notification workflows.
6. Export raw JSON for key alerts to make evidence stronger in the final report.
7. Create a network diagram showing Windows, Ubuntu, Wazuh Server, and VirtualBox host-only network.

7.3 Key Takeaway

This lab successfully demonstrates the core workflow of a Tier 1 SOC analyst: deploy monitoring, ingest endpoint logs, generate controlled attack telemetry, identify alerts, map activity to MITRE ATT&CK, triage incidents, and document recommended response actions. The project shows hands-on experience with Wazuh SIEM/XDR, Windows and Linux endpoint monitoring, Sysmon, FIM, vulnerability detection, and alert investigation.

